DEVELOPMENT & DEPLOYMENT OF AI IN LIVER TRANSPLANTATION

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January 28th, 2026
MSc LMP 2392H course

Medicine
UNIVERSITY OF TORONTO

Transplant AI Initiative

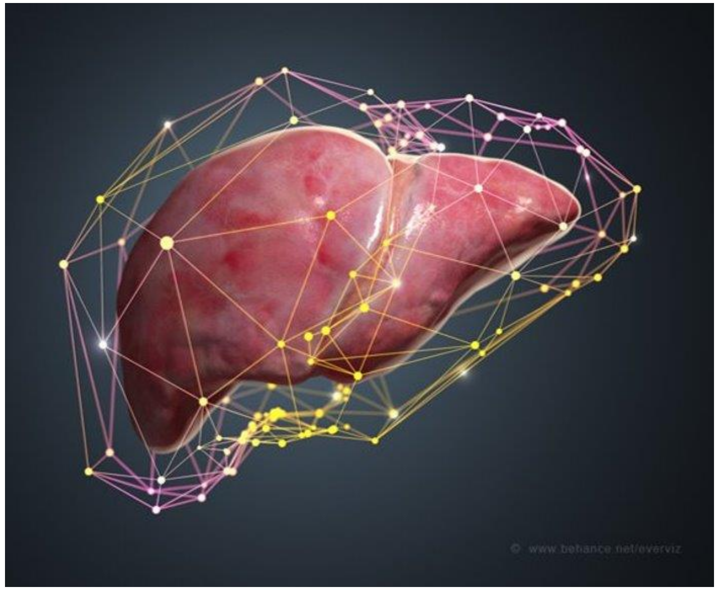
UHN Ajmera Transplant Centre

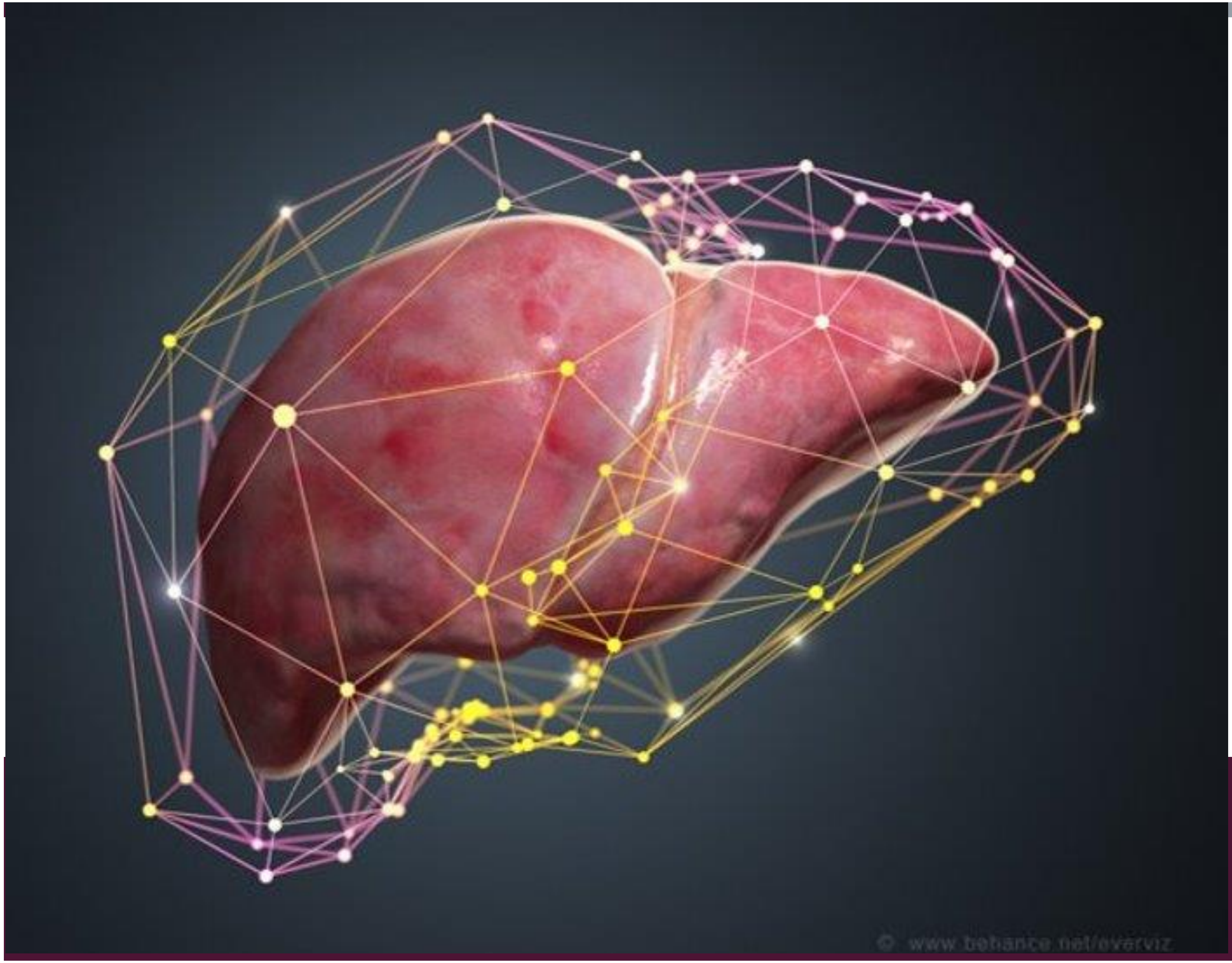
DISCLOSURES

- Novo Nordisk, Knight, CareDx, Natera, Oncoustics, Roche, Merck, Astra Zeneca
- **Peer-Reviewed Grants:** CIHR, Terry Fox Research institute, NSERC, ERA – Govt of Ontario, New Frontiers Research Fund, BBDC, CDTRP, Stem cell network, UHN Foundation

OUTLINE

- Why AI in Transplantation?
- Optimizing equity in candidacy and waitlist prioritization
- Optimizing Post-Transplant Patient Survival
- Optimizing Graft Survival in Liver Transplant Recipients

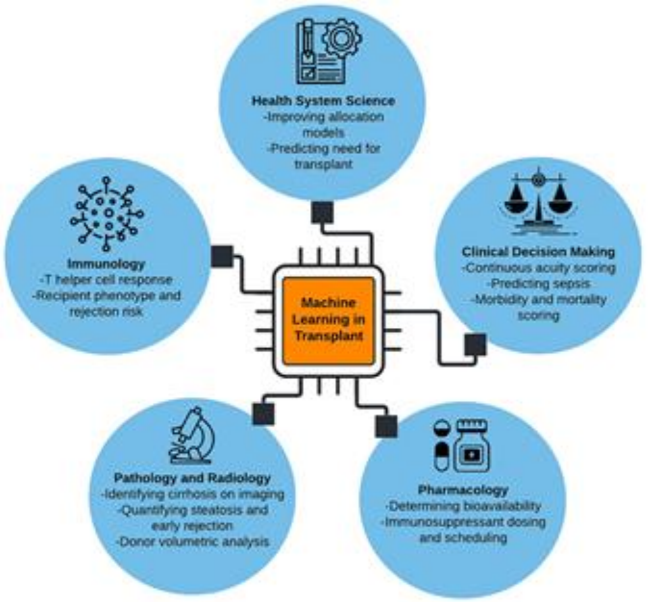




WHY AI IN LIVER TRANSPLANT?

WHY ML IN LIVER TRANSPLANT?

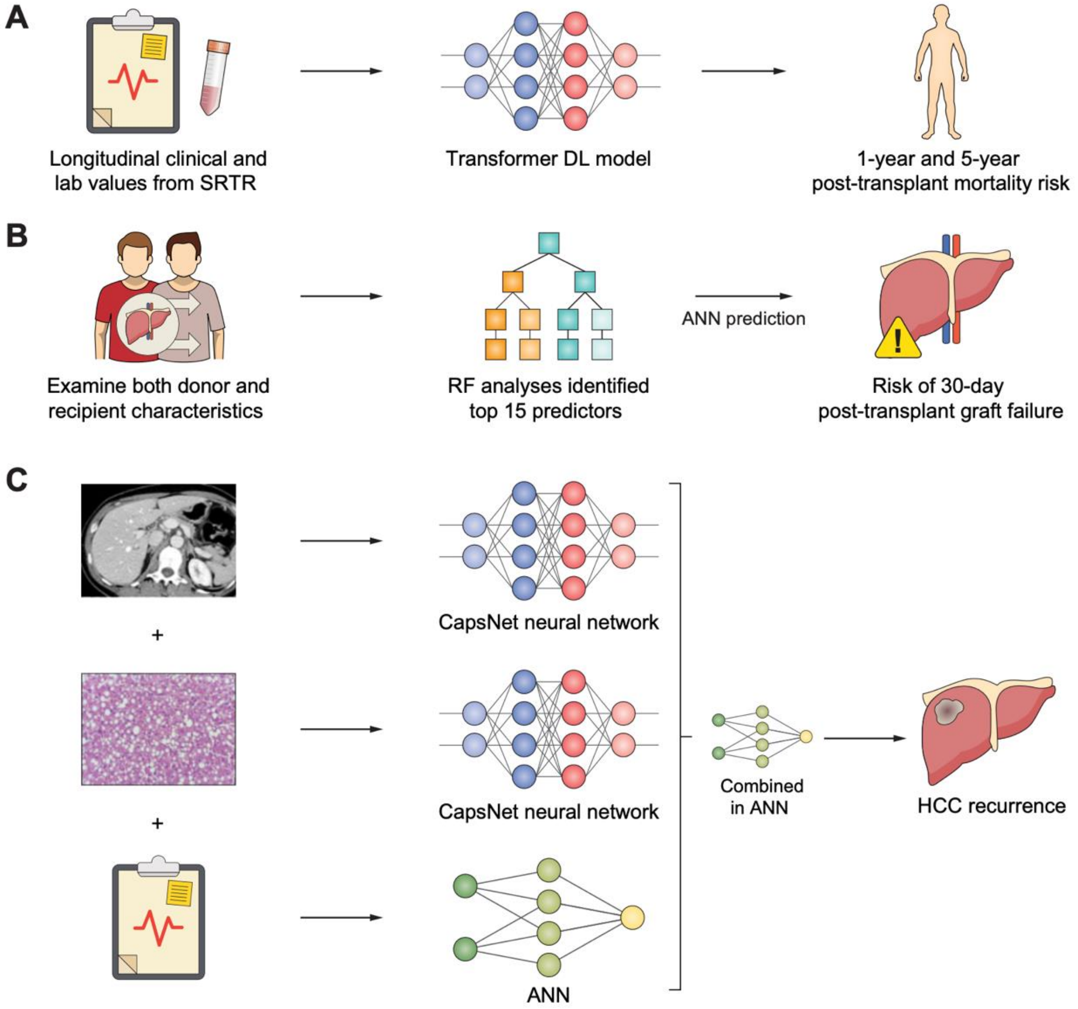
- Complex patient population
- Clinical variables: sex, ethnicity, genetics, environmental exposures, lifestyle, BMI, diabetes, immunosuppression
- Complex, non-linear patterns in liver enzymes, liver function tests, platelets, creatinine, sodium
- Electronic health record data, imaging technologies, histology, wearables, molecular data



[1]Spann A, Strauss A, Bhat M. The Role of Artificial Intelligence in Chronic Liver Diseases and Liver Transplantation. 2025 Aug;169(3):456-470. Gastroenterology
[2]Spann A, Wang B, Goldenberg A, Bhat M. Applying Machine Learning in Liver Disease & Transplantation: A Comprehensive Review, Hepatology 2020
[2]Gotlieb N, Spann A, Wang B, Bhat M. Applying Machine Learning in Solid Organ Transplantation. NPJ Digital Medicine, 2022
[3]Tran J, Bhat M. Machine Learning in Liver Transplantation. Hepatology International 2022.

BRINGING OBJECTIVITY TO COMPLEX DECISIONS IN TRANSPLANT

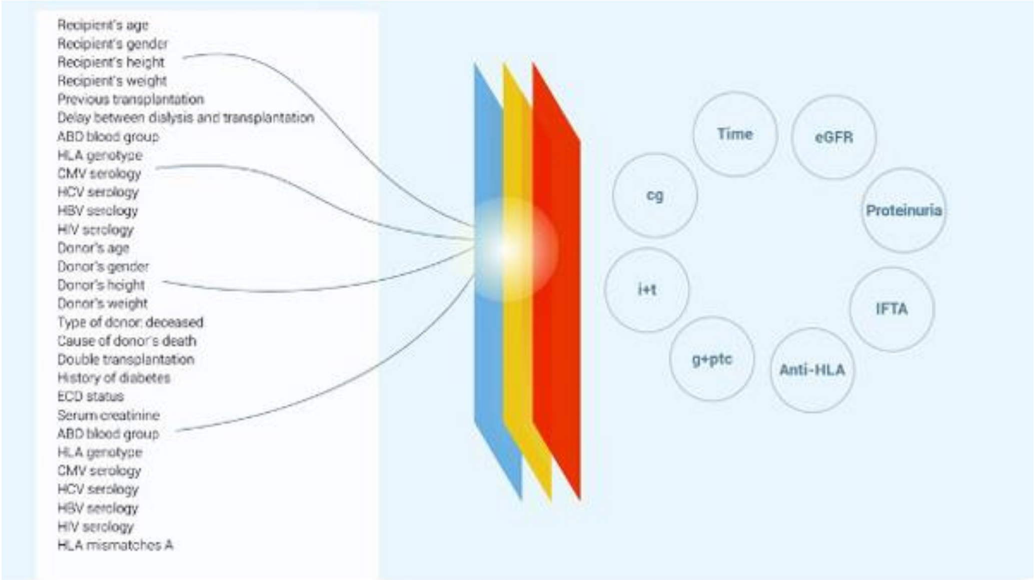
- Using AI as Co-Pilot for **Complex Decision**
- Identifying Suitable Donor Organs
- Waitlist prioritization
- Applications in Donor-recipient matching
- Prediction of Outcomes



Potential to bring objectivity to decision-making – though if data biased, risk of decision being biased

Bhat M, Rabindranath M, Sordid-Chara B, Simonetto D. Journal of Hepatology 2023

IBOX SCORING SYSTEM FOR KIDNEY GRAFT SURVIVAL



Prediction system for risk of allograft loss in patients receiving kidney transplants: international derivation and validation study

Alexandre Loupy,[1,2] Olivier Aubert,[1,2] Babak J Orandi,[3] Maarten Naesens,[4] Yassine Bouatou,[1] Marc Raynaud,[1] Gillian Divard,[1] Annette M Jackson,[5] Denis Viglietti,[1,6] Magali Giral,[7] Nassim Kamar,[8] Olivier Thaunat,[9] Emmanuel Morelon,[9] Michel Delahousse,[10] Dirk Kuypers,[4] Alexandre Hertig,[11] Eric Rondeau,[11] Elodie Bailly,[11] Farsad Eskandary,[12] Georg Böhmig,[12] Gaurav Gupta,[13] Denis Glotz,[1,6] Christophe Legendre,[1,2] Robert A Montgomery,[14] Mark D Stegall,[15] Jean-Philippe Empana,[1,16] Xavier Jouven,[1] Dorry L Segev,[17] Carmen Lefaucheur[1,6]

- Considered several patient, donor, and graft parameters
- Has become clinical trial endpoint in kidney transplant

Loupy et al, BMJ 2019

MACHINE LEARNING & EVLP: THE INSIGHTX MODEL

- **The InsighTx AI Model for EVLP:**
- First machine-learning model for ex vivo organs
- More accurately assesses organ health
- Predicts post-transplant recipient outcomes
- Standardize the lung transplant field

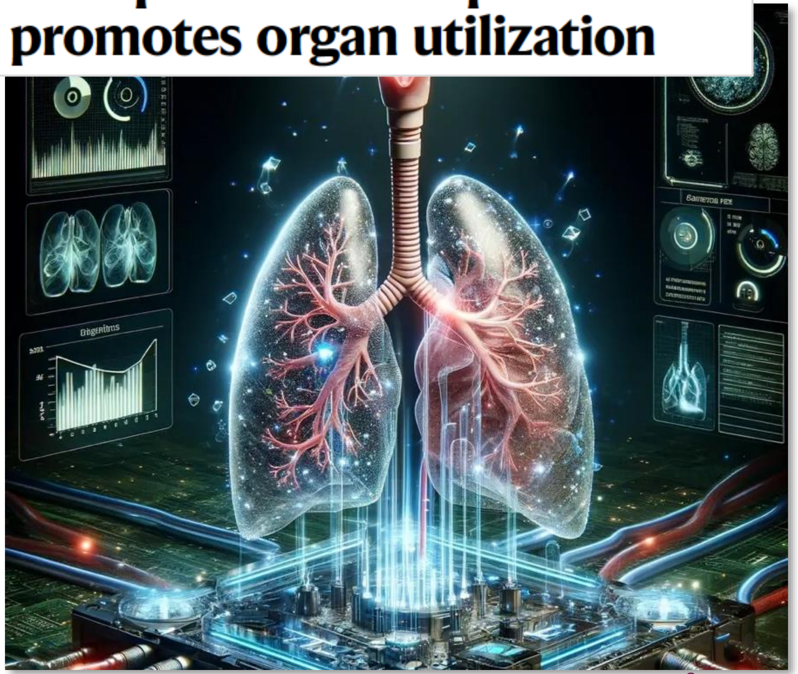
- **InsighTx contains donor and EVLP features**
- Donor features (i.e., age, sex, BMI…)
- Biological assessments (i.e., IL-6, IL-8…)
- Physiological assessments (i.e., compliance, PO_2…)
- Biochemical assessments (i.e., pH, glucose…)

- **There are over 91 different features of a lung on an ex vivo circuit that are**
- **looked at simultaneously using InsighTx**

nature communications

Article https://doi.org/10.1038/s41467-023-40468-7

A machine-learning approach to human ex vivo lung perfusion predicts transplantation outcomes and promotes organ utilization



Sage/Keshavjee et al, *Nature Communications* 2023

INSIGHTX ACCURATELY PREDICTS EVLP AND TX

Model Performance

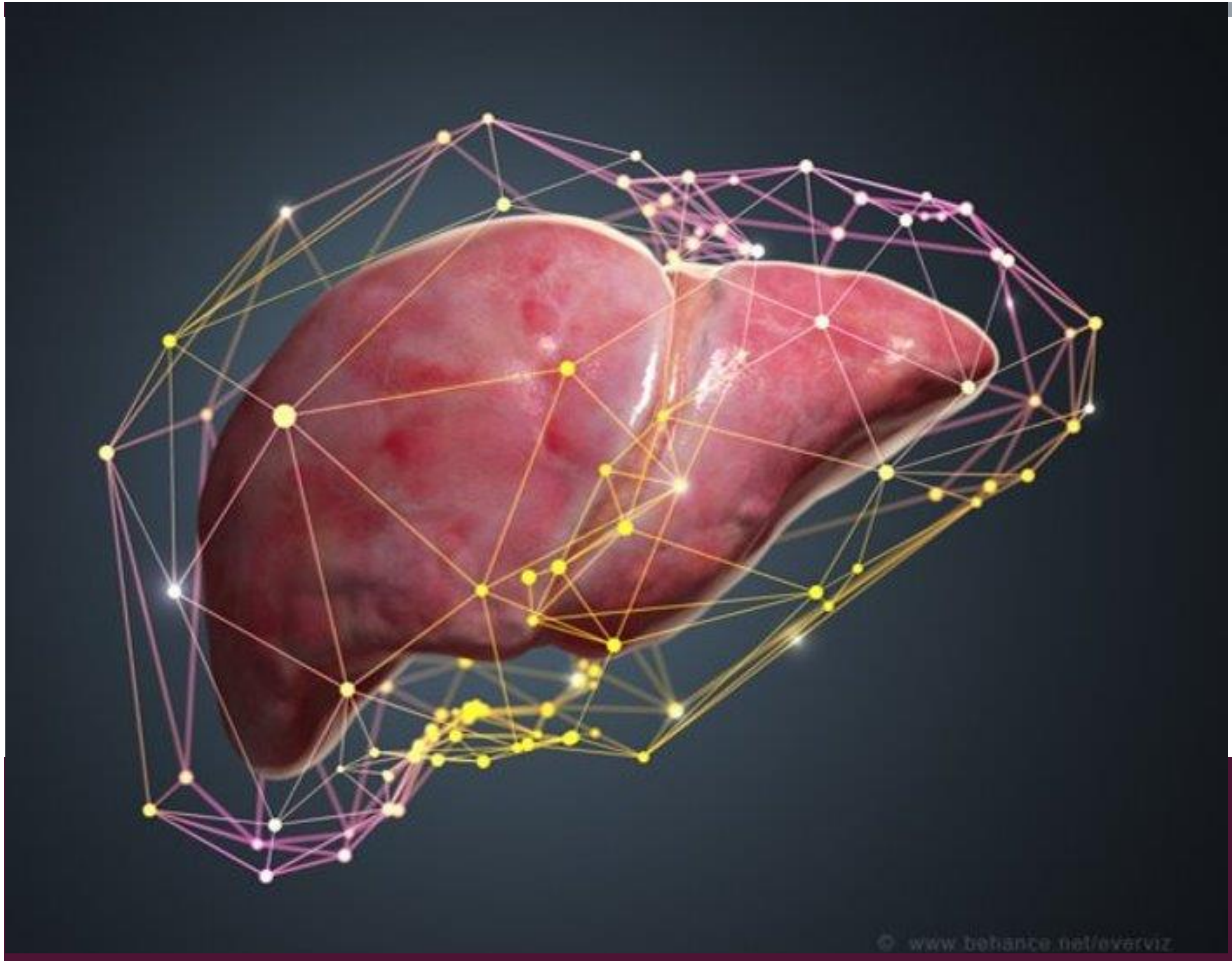
InsighTx	Number of Cases	AUROC
Training Dataset	504	79±3%
Test Set 1	97	75±4%
Test Set 2	124	85±3%

A safe increase in LTx using InsighTx

	Transplant Decision OR [95% CI]	Clinical Impression of Donor Lung Δ [95% CI]
Suitable Donor Lungs (n=12 lungs)	13 [95% CI: 4 to 45]	+0.95 [95% CI: 0.4 to 1.51]
Unsuitable Donor Lungs (n=8 lungs)	0.4 [95% CI: 0.16 to 0.98]	-0.31 [95% CI: -0.75 to 0.14]

AUROC=Area Under Receiver Operating Characteristic Curve

Sage/Keshavjee et al, *Nature Communications* 2023

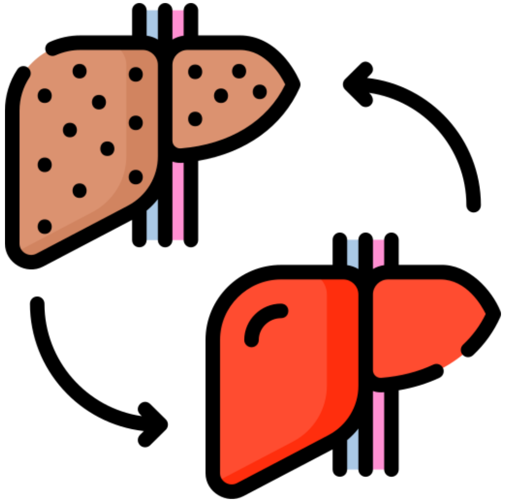


OPTIMIZING EQUITY IN CANDIDACY AND WAITLIST PRIORITIZATION

THE AI AGENT IN THE ROOM: INFORMING OBJECTIVE DECISION MAKING AT THE TRANSPLANT SELECTION COMMITTEE

Liver Transplantation (LT) is the only definitive treatment for end stage liver disease

- Decided by a Multidisciplinary LT Selection Committee
- Process of selecting candidates for transplant is vastly understudied, prone to[1,2]:
  - Inconsistency
  - Center-specific rules (written or unwritten)
  - External pressures
  - Hidden biases against select psychosocial profiles
  - Recency bias



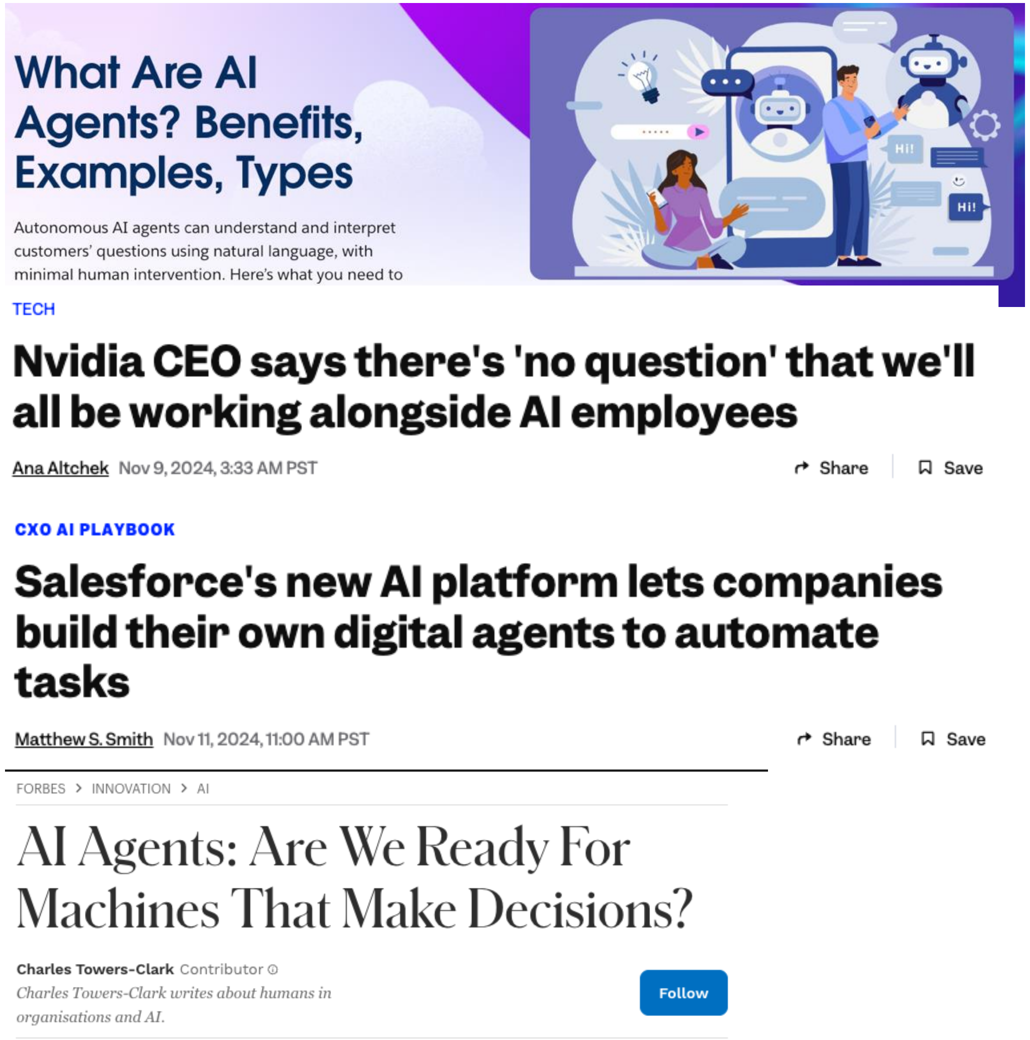
1. Volk et al. (2011) Ann Int Med
2. Martin et al. (2014) Hepatology

INTRODUCTION: AI AGENTS

- Large language models that use AI to perform tasks, interact with their environment, and learn from experience.
- Can engage in **inter-agent communication to collaborate**, share information, and make decisions collectively to solve problems.

In the business world, AI agents are used for:

- Increasing efficiency, automation
- Process improvement initiatives



What Are AI Agents? Benefits, Examples, Types

Autonomous AI agents can understand and interpret customers' questions using natural language, with minimal human intervention. Here's what you need to

TECH

Nvidia CEO says there's 'no question' that we'll all be working alongside AI employees

Ana Altchek Nov 9, 2024, 3:33 AM PST Share Save

CXO AI PLAYBOOK

Salesforce's new AI platform lets companies build their own digital agents to automate tasks

Matthew S. Smith Nov 11, 2024, 11:00 AM PST Share Save

FORBES > INNOVATION > AI

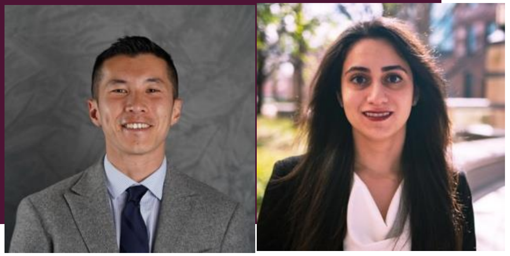
AI Agents: Are We Ready For Machines That Make Decisions?

Charles Towers-Clark Contributor

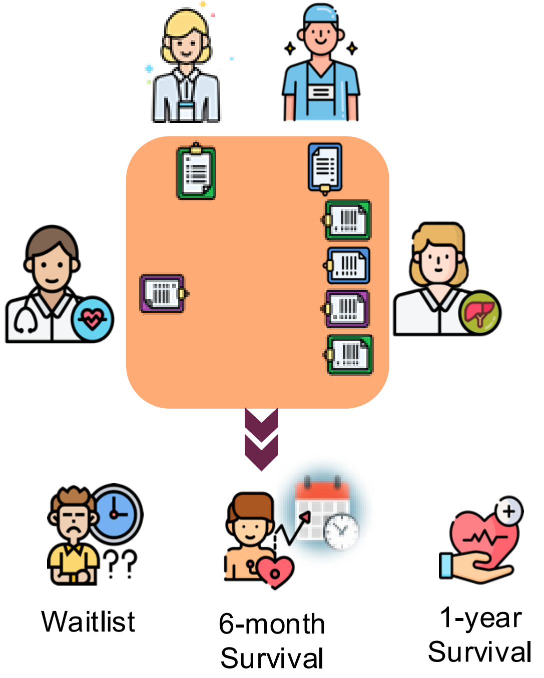
Charles Towers-Clark writes about humans in organisations and AI.

Follow

MULTI AI AGENTS: SIMULATING THE TRANSPLANT SELECTION COMMITTEE



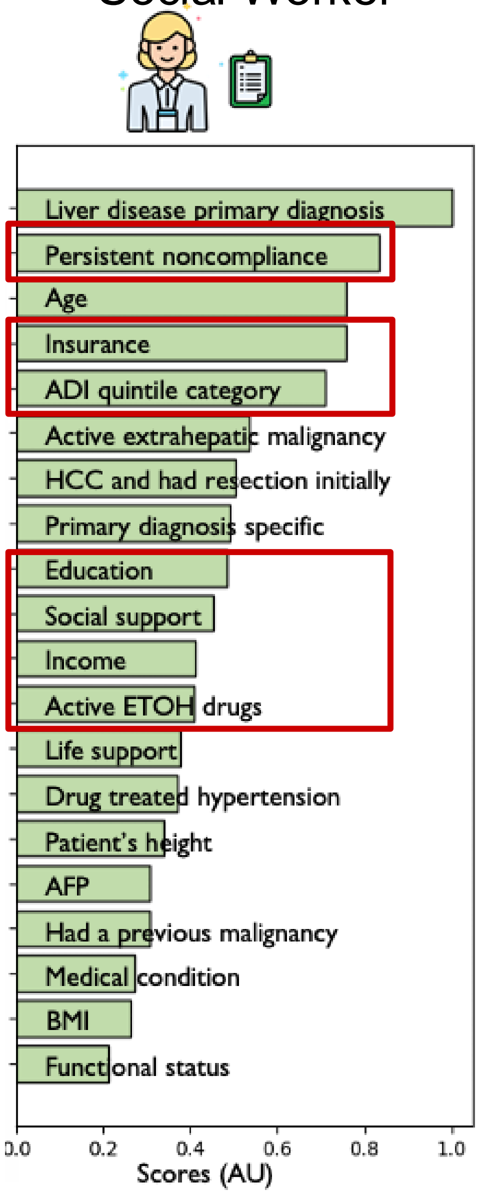
- AI agents can be used to simulate LT Selection committee
- Large Language Model representing each subspecialist on SC
- AI-SC reviewed LT recipients in SRTR (N=7,033) and generated cases with standard absolute contraindications to LT (N=1,379).
- AI-SC can accurately waitlist LT candidates (98.2%) and predict 6-month (94.8%) and 1-year (92.0%) survival.

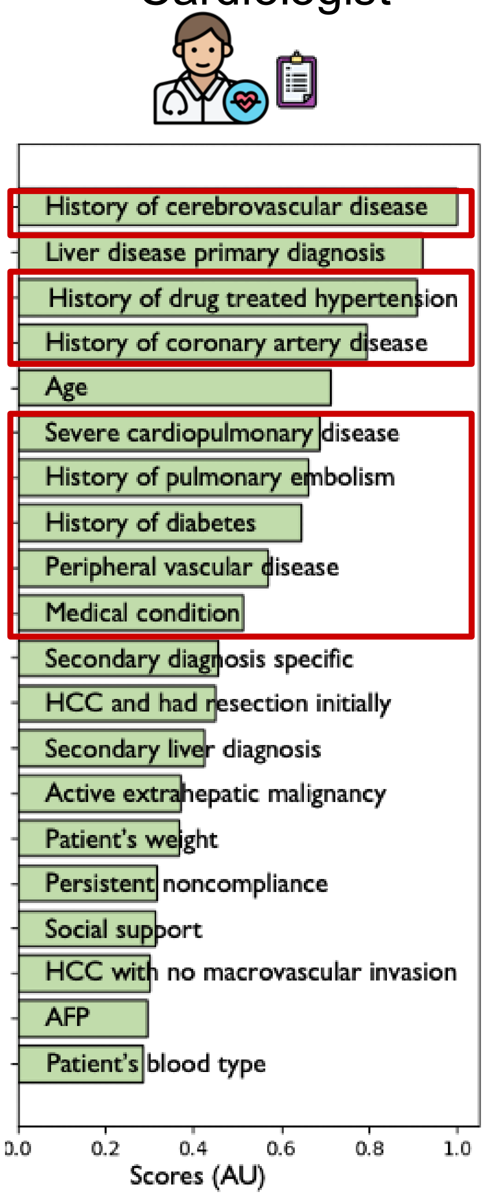


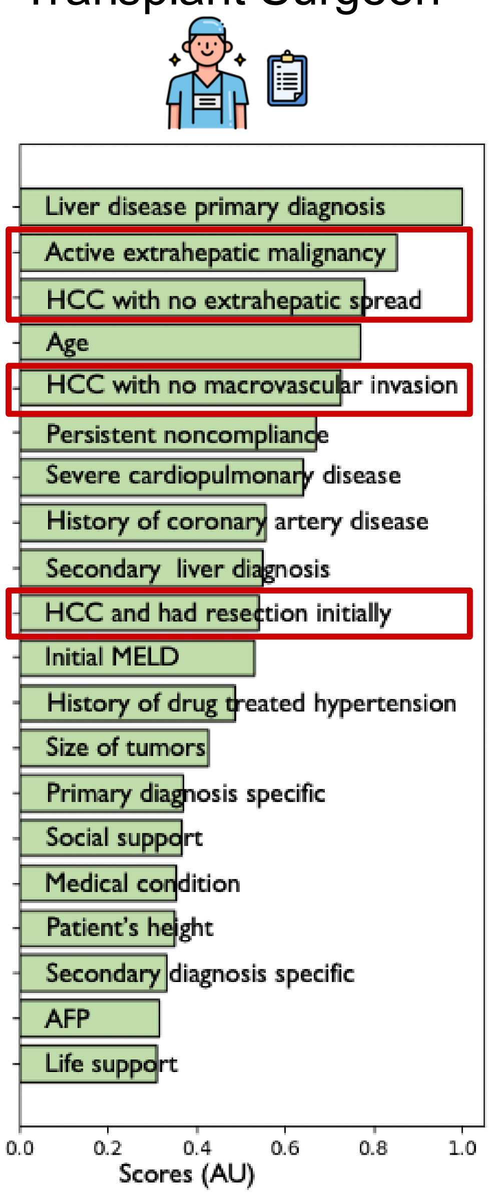
	Accuracy (95%CI)	Sensitivity (95%CI)	Specificity (95%CI)	Precision (95%CI)	F1-score (95%CI)
Contraindications to LT	98.19% (97.90%-98.44%)	1.00 (0.99-1.00)	0.91 (0.89-0.92)	0.98 (0.98-0.99)	0.99 (0.99-0.99)
6-month survival	94.88% (94.37%-95.29%)	1.00 (0.99-1.00)	0.75 (0.73-0.77)	0.94 (0.94-0.95)	0.97 (0.97-0.97)
1-year survival	92.00% (91.43%-92.58%)	1.00 (0.99-1.00)	0.66 (0.64-0.68)	0.91 (0.90-0.92)	0.95 (0.95-0.95)

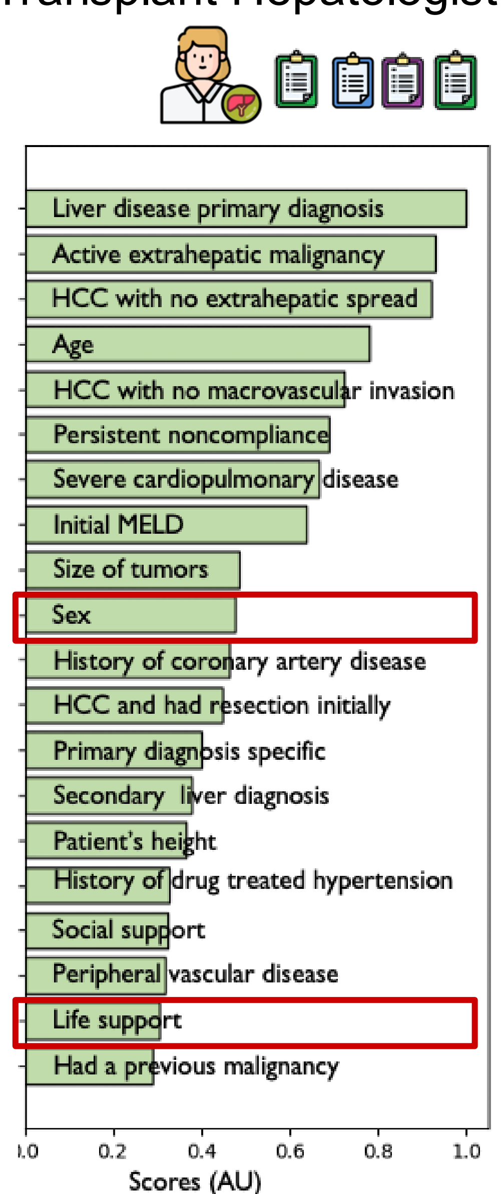
Hasjim B, Azarfar G, Sharma D, Bhat M. Accepted in Lancet Digital Health

CHARACTERISTICS ACCOUNTING FOR EACH AGENT'S DECISION









Hasjim B, Azarfar G, Sharma D, Bhat M. under final revisions, Lancet Digital Health

DASHBOARD

AI Committee

Light

AI Committee Analysis Results

NEW ANALYSIS

Patient's Vignette

Jane doe is a **70-year-old** woman with decompensated alcohol-related cirrhosis (ALD), and a BMI of 21.9 presents for transplant assessment. She undergoes weekly large volume paracentesis (6L) and has a current MELD-Na score of 19. Initial concerns regarding frailty and sarcopenia have improved; she is now independently mobile.

Past Medical History Past Medical History includes breast cancer in remission since 2017 (treated with lumpectomy and chemoradiation). There is no formal diagnosis of alcohol use disorder; she has been cleared by psychiatry and has had two negative urine ETG tests. She has family and financial support with a living donor.

Clinical Workup

MELD

MELD 3.0: 17
Creatinine (mg/dL): 140
Bilirubin (mg/dL):
INR: 1.5

Cardiac Workup

ECG: Normal sinus rhythm, HR 60
Echo: LVEF 60-65%, moderate left atrial dilatation, RVSP not measurable
Stress test: No evidence of ischemia, preserved systolic function
CT Chest: Moderate LAD coronary atherosclerosis

GI & Imaging

EGD: No varices, portal hypertensive gastropathy
Colonoscopy: Normal
Chest CT: Normal
Abdominal CT: Further Doppler imaging recommended due to elevated creatinine
Abdominal Ultrasound: Cirrhotic liver, patent portal vein, splenomegaly

Serology

HBV: HBsAB negative, HBsAg negative, HBcAb negative

AI Committee Decision

The patient is a 70-year-old female with decompensated alcohol-associated cirrhosis (ALD), MELD 3.0 score of 17, primarily elevated due to creatinine (140 µmol/L) and INR (1.5). She exhibits sarcopenia and is at high risk of waitlist dropout. Following evaluation by Dr.John Smith, she is not a candidate for TIPS. Additionally, concerns have been raised about non-adherence to a low-sodium diet. Given these factors and absence of a potential living donor, the committee recommends conservative management at this time. A liver transplant will not be offered unless there is marked improvement in ascites control with strict dietary adherence.

Final Recommendation
Conservative management. A liver transplant will not be offered at this time.

Contraindications

Absolute Contraindication

The patient has no absolute contraindications.

Relative Contraindication

Considering their functional status, age is a relative contraindication.

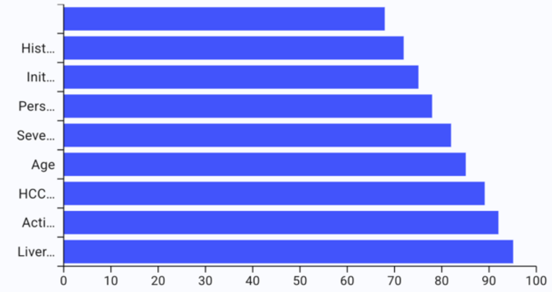
Individual Analysis

TRANSPLANT HEPATOLOGIST | ANESTHESIOLOGIST | CARDIOLOGIST | TRANSPLANT SURGEON | SOCIAL WORKER

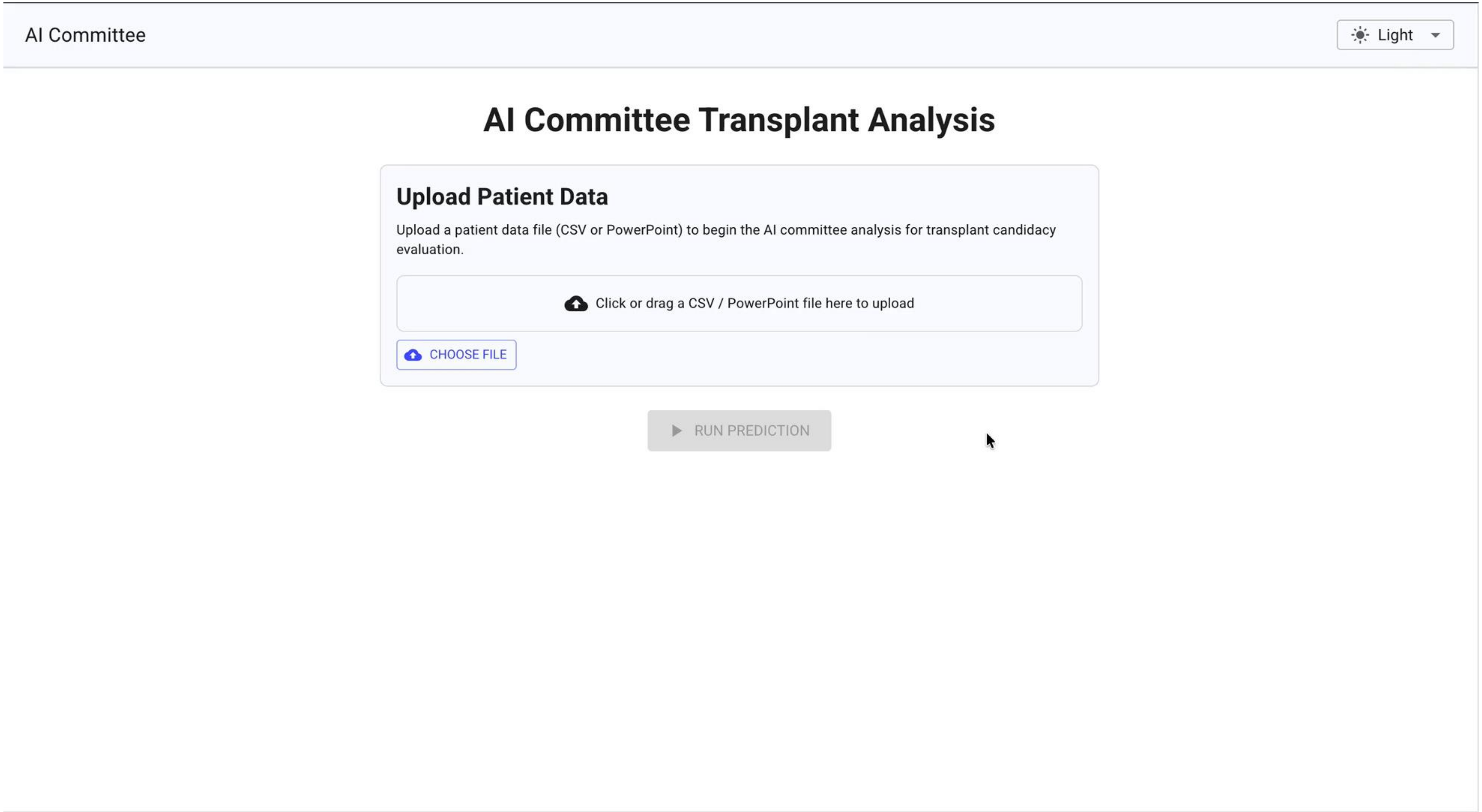
Social Worker's Report

The social worker confirms the patient has strong family and financial support. There is no formal diagnosis of alcohol use disorder, and the patient has been compliant with ETG testing. Adherence to diet is a concern.

Social Worker's Score Analysis



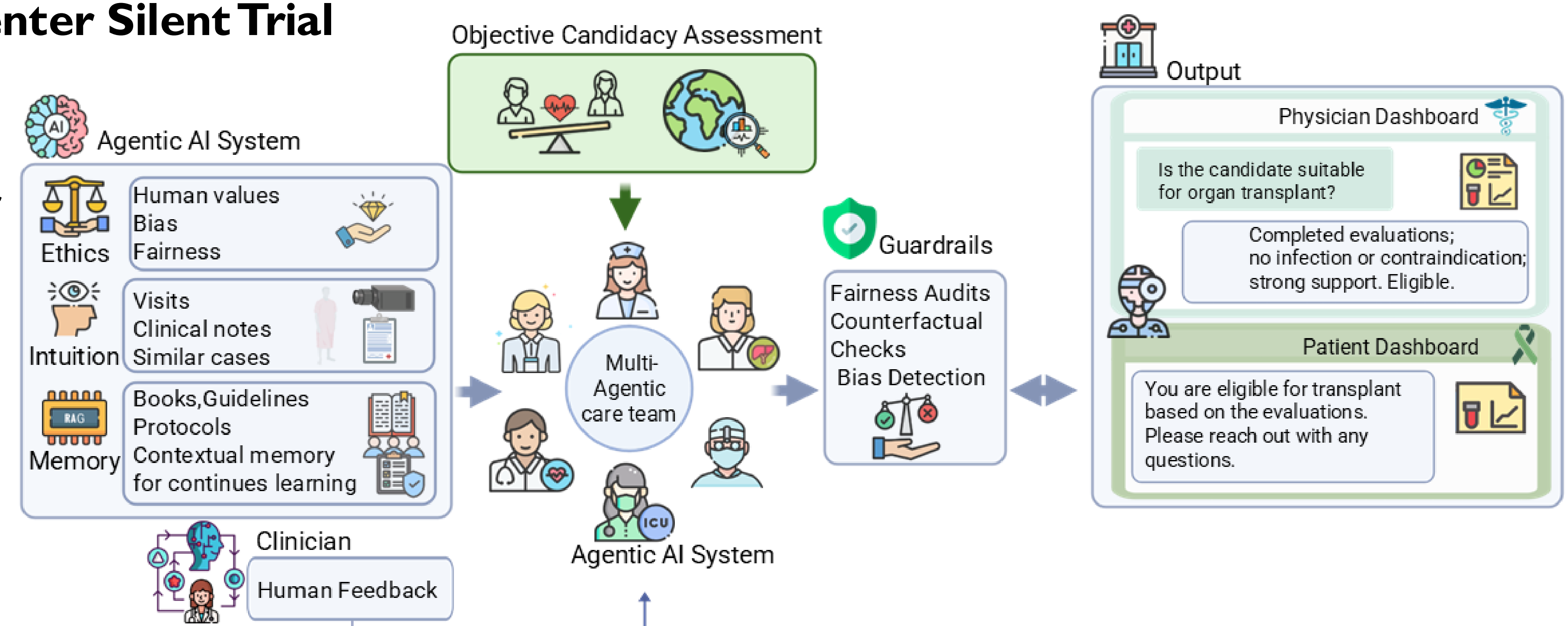
DASHBOARD


AI Committee
Light
AI Committee Transplant Analysis
Upload Patient Data
Upload a patient data file (CSV or PowerPoint) to begin the AI committee analysis for transplant candidacy evaluation.
Click or drag a CSV / PowerPoint file here to upload
CHOOSE FILE
RUN PREDICTION

SILENT TRIAL

Schematic overview of the multi agent transplant selection committee managed a by physician-facing agent that, upon request, activates the agentic transplant team by assembling relevant specialist agents, and adjusting dashboards individualized to clinician and patient.

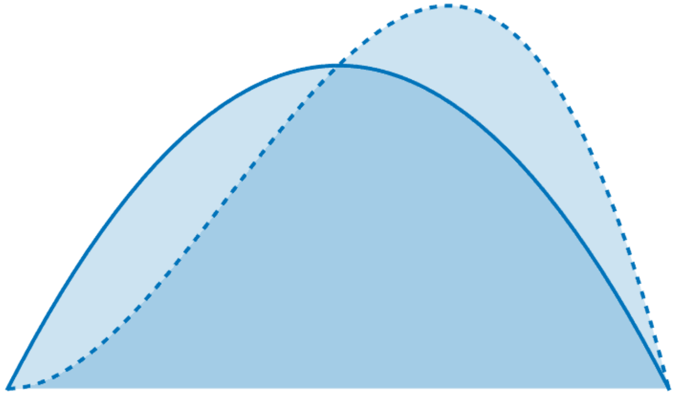
- **6 months** of **Multi Center Silent Trial**
  - Transplant Coordinator
  - Hepatologist
  - Social Worker
  - Cardiologist
  - Surgeon
  - Anesthesiologist
  - Psychiatrist



Inequities in Liver Transplant Allocation

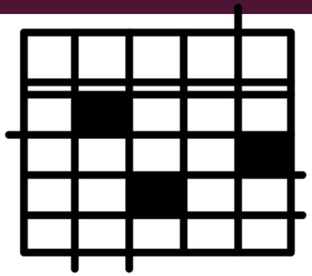
Disadvantages Female Patients (Locke et al., 2020; Karnam et al, 2021)

Disadvantages Patients w/ PBC (Singal et al., 2021)



Shift in Underlying Patient Cohort (HCV to ALD/MASH)

DYNAMELD: DYNAMIC MODEL OF END-STAGE LIVER DISEASE FOR FAIR RISK ASSESSMENT

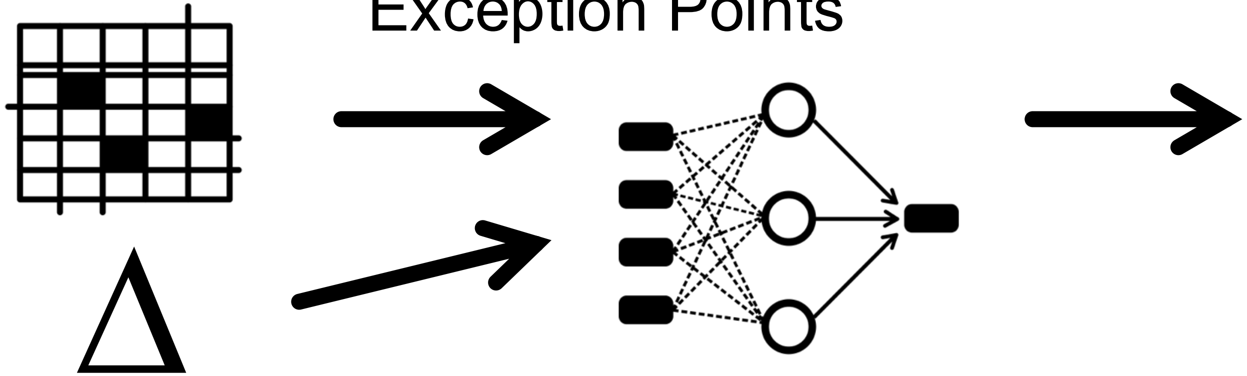


53,046 Patients Listed for Transplant in the United States since 2016

Adult Patients with Decompensated Cirrhosis and Without Exception Points

13 Static and Time-Varying Covariates per Patient

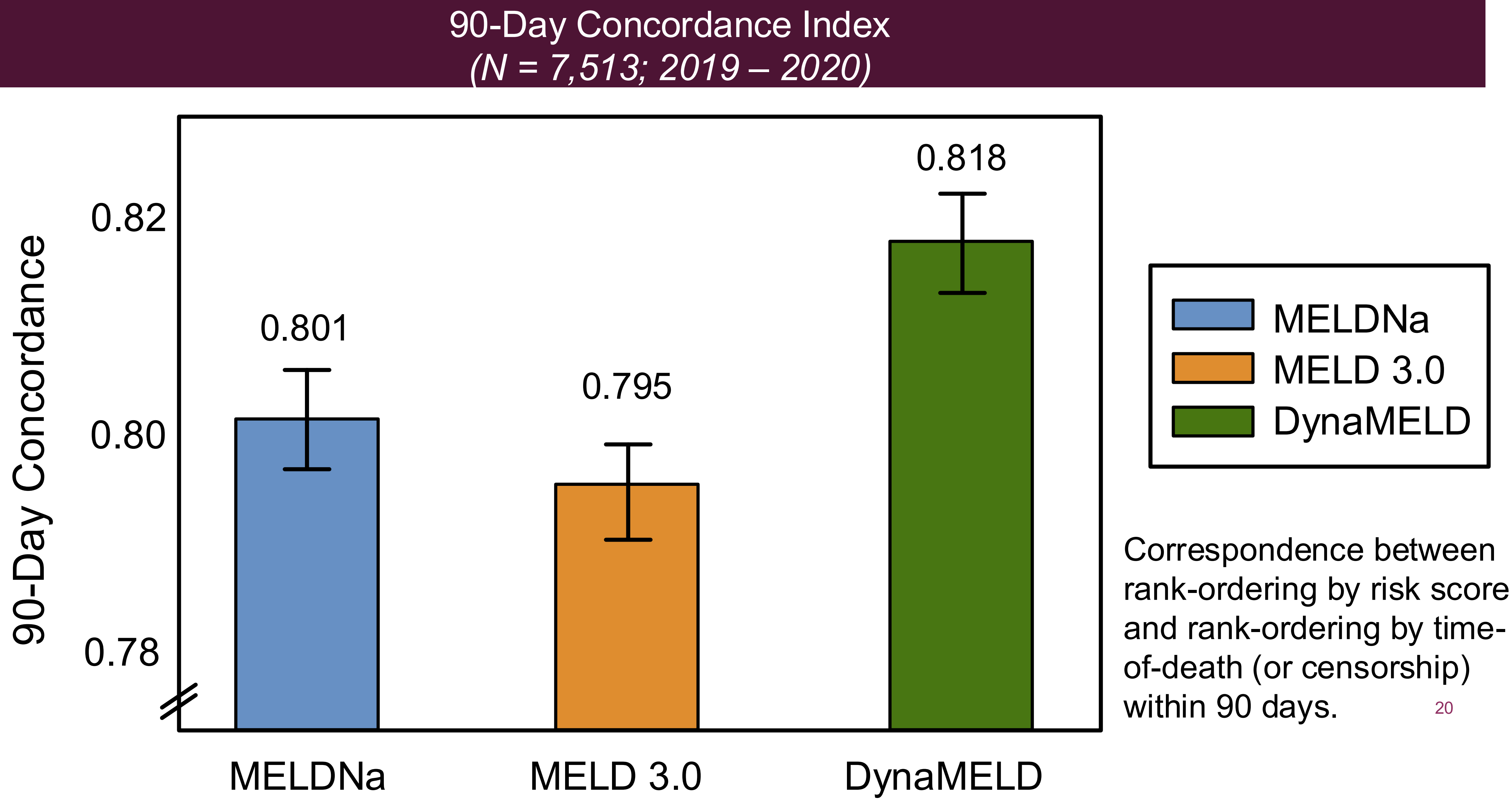
DeepSurv:
(Nonlinear Survival Analysis)



Patient Risk Score

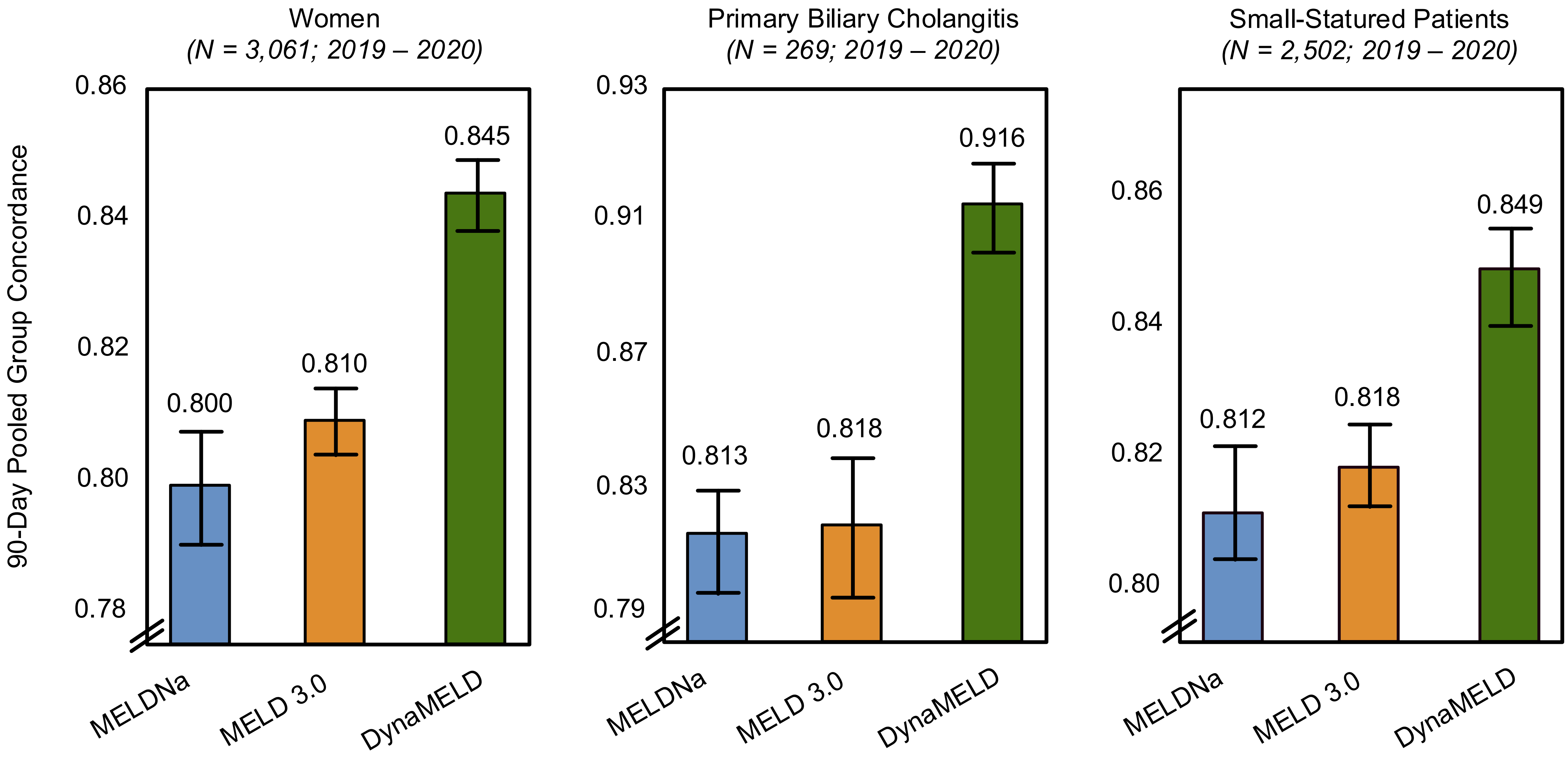
Cooper M, Krishnan R, Bhat M.
https://www.medrxiv.org/content/10.1101/2024.11.19.24316852v

CONCORDANCE ON HELD-OUT VALIDATION DATA

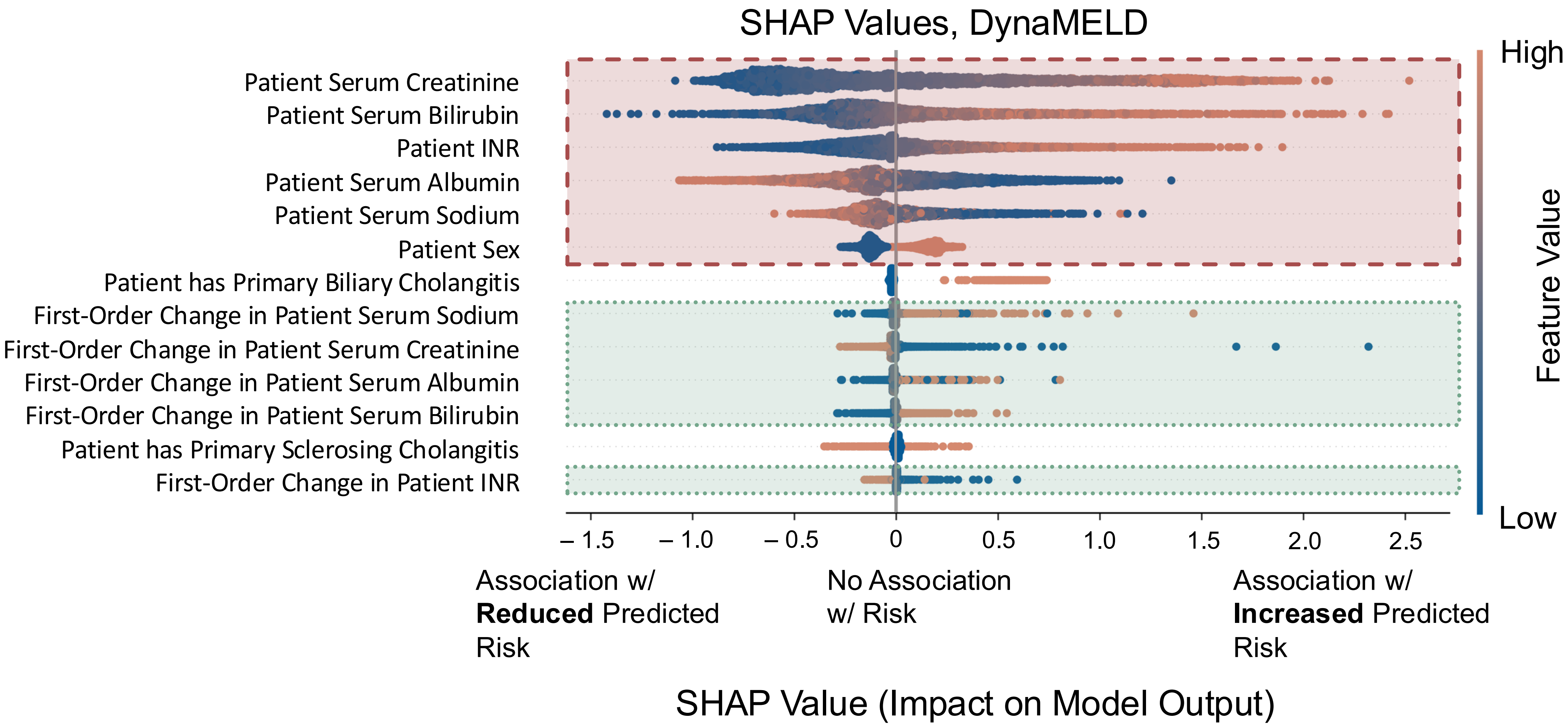


Correspondence between rank-ordering by risk score, and rank-ordering by time-of-death (or censorship) within 90 days.

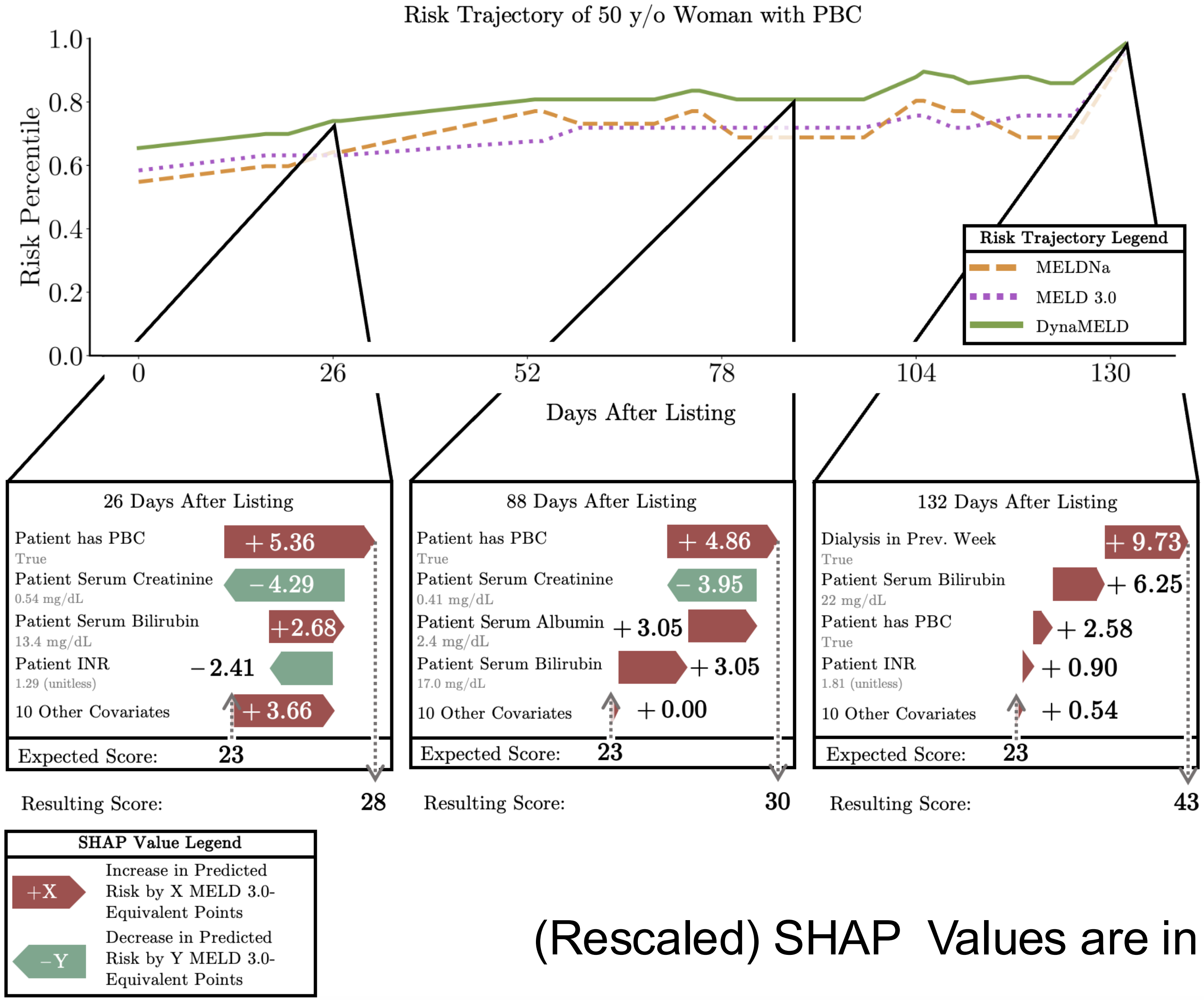
POOLED GROUP CONCORDANCE



INTERPRETABILITY: SHAPLEY ANALYSIS

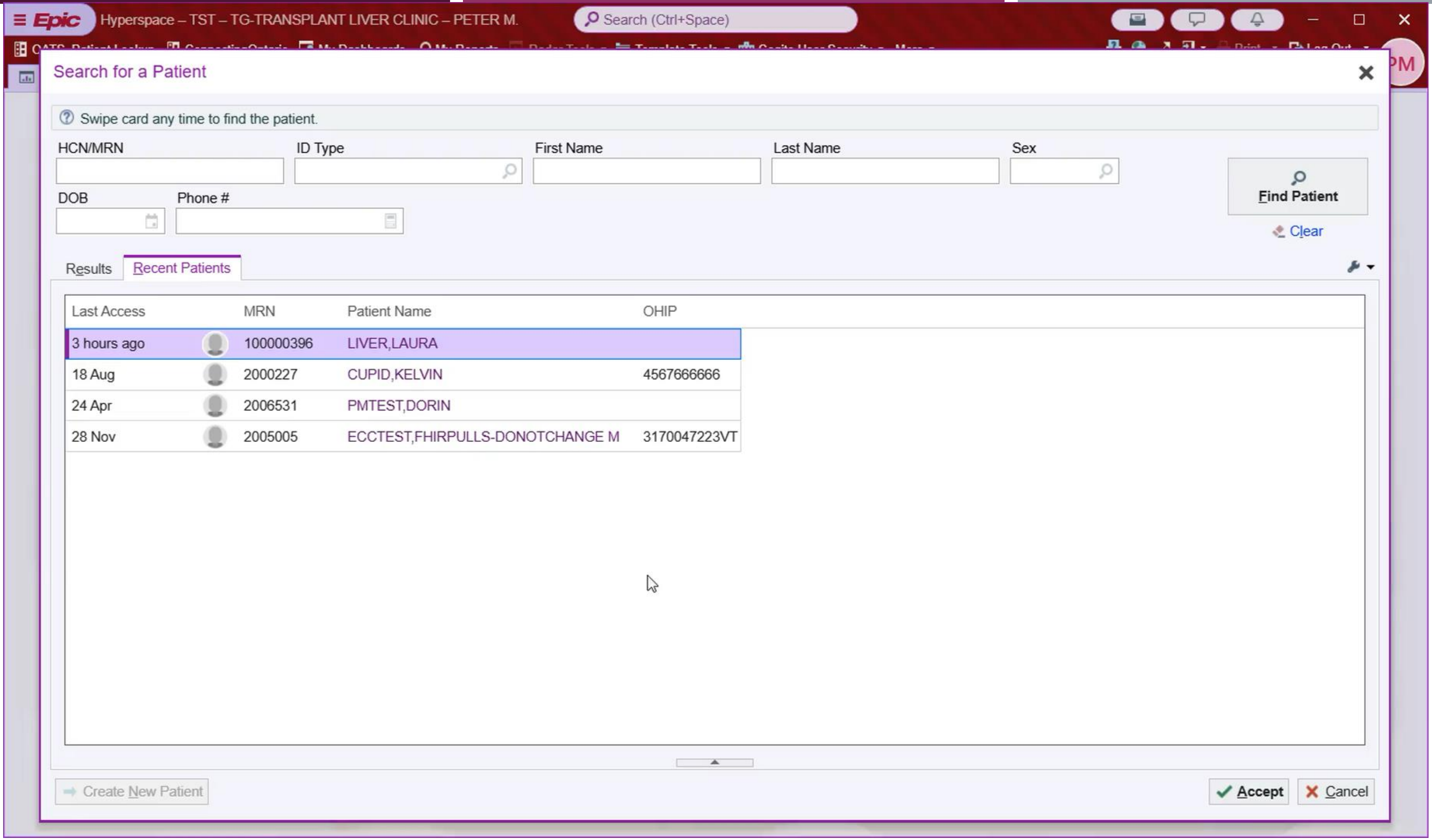


Individualized Interpretations of Dynamic Trajectories



(Rescaled) SHAP Values are in Units of MELD Points

DynaMELD dashboard



DYNAMELD RECLASSIFIES DISADVANTAGED GROUPS INTO HIGHER RISK TIERS

We simulated a silent trial on 381 patients listed for liver transplant at the University Health Network (June 2022–November 2024)

Identified Sub-Group, *N*	Number Up-Ranked vs. MELD 3.0	Number Down-Ranked vs. MELD 3.0	Difference, *N* (%)
Decedents, *N* = 103	20 (13, 28)	15 (8, 23)	5 (-6, 17), 4.85%
Women, *N* = 149	27 (18, 36)	23 (15, 31)	4 (-9, 18), 2.68%
Patients w/ PSC, *N* = 36	13 (8, 19)	4 (1, 8)	9 (2, 16), 25.0%
Patients w/ PBC, *N* = 16	9 (5, 13)	0 (0, 0)	9 (5, 13), 56.25%

THE LANCET
Digital Health

Development and validation of a machine-learning model to reduce futile procurements in donations after circulatory death in liver transplantation in the USA: a multicentre

Background:

- ❑ High rates of attempted but subsequently terminated (futile)procurements
- ❑ Significant resource waste and operational burden

Aim:

- ❑ Develop and validate a machine-learning model to predict progression to death in DCD donors

Method:

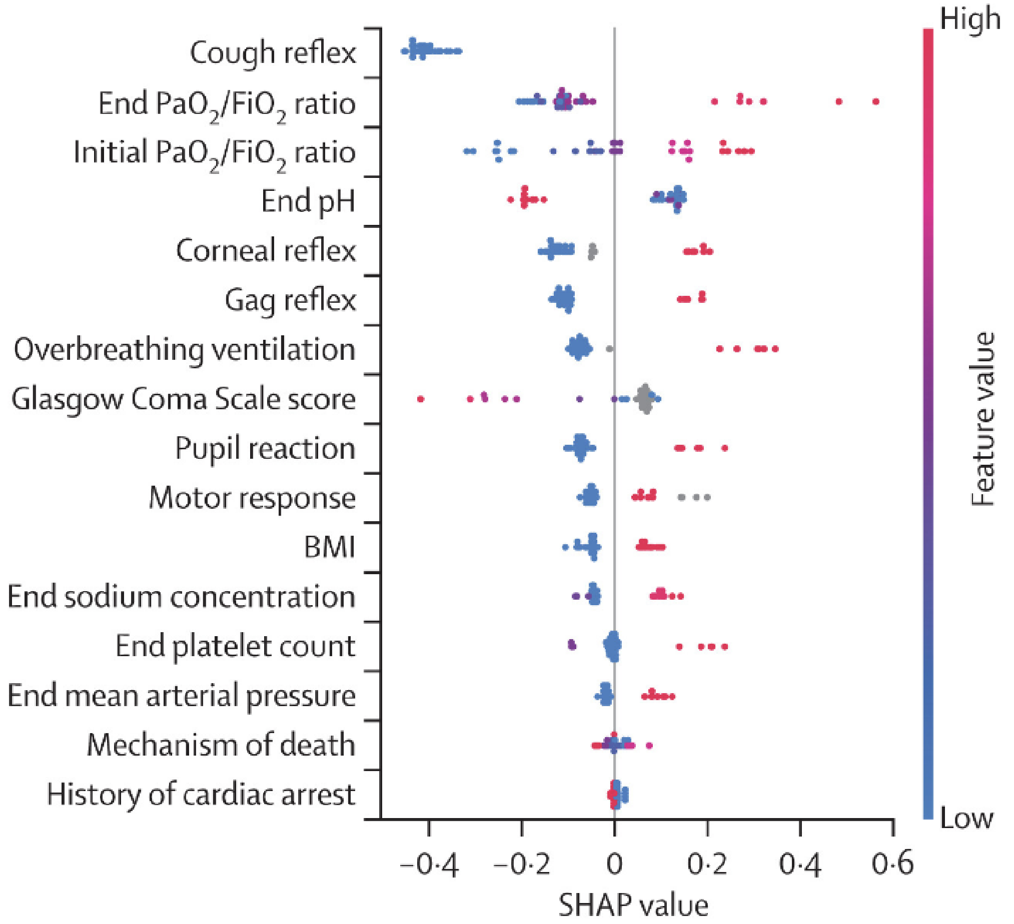
- ❑ Data from 6 centers in the USA
  - o Training set: 1616 donors from Dec.2022-Jun.2023
  - o Retrospective validation set: 398 donors from Jul-Aug.2023
  - o Prospective validation set: 207 donors from Mar-Sep.2024
- Input features: neurological, biochemical, respiratory, and circulatory parameters
- Outcome: Death at 30, 45, 60 min post-extubation
- Models: Light Gradient Boosting Machine (LightGBM)random forest, eXtreme Gradient Boosting (XGBoost),Categorical Boosting (CatBoost),neural networks (multilayer perceptron), and logistic regression

Yanagawa R, Iwadoh K, Nakayama T, et al. The Lancet Digital Health, Nov.20

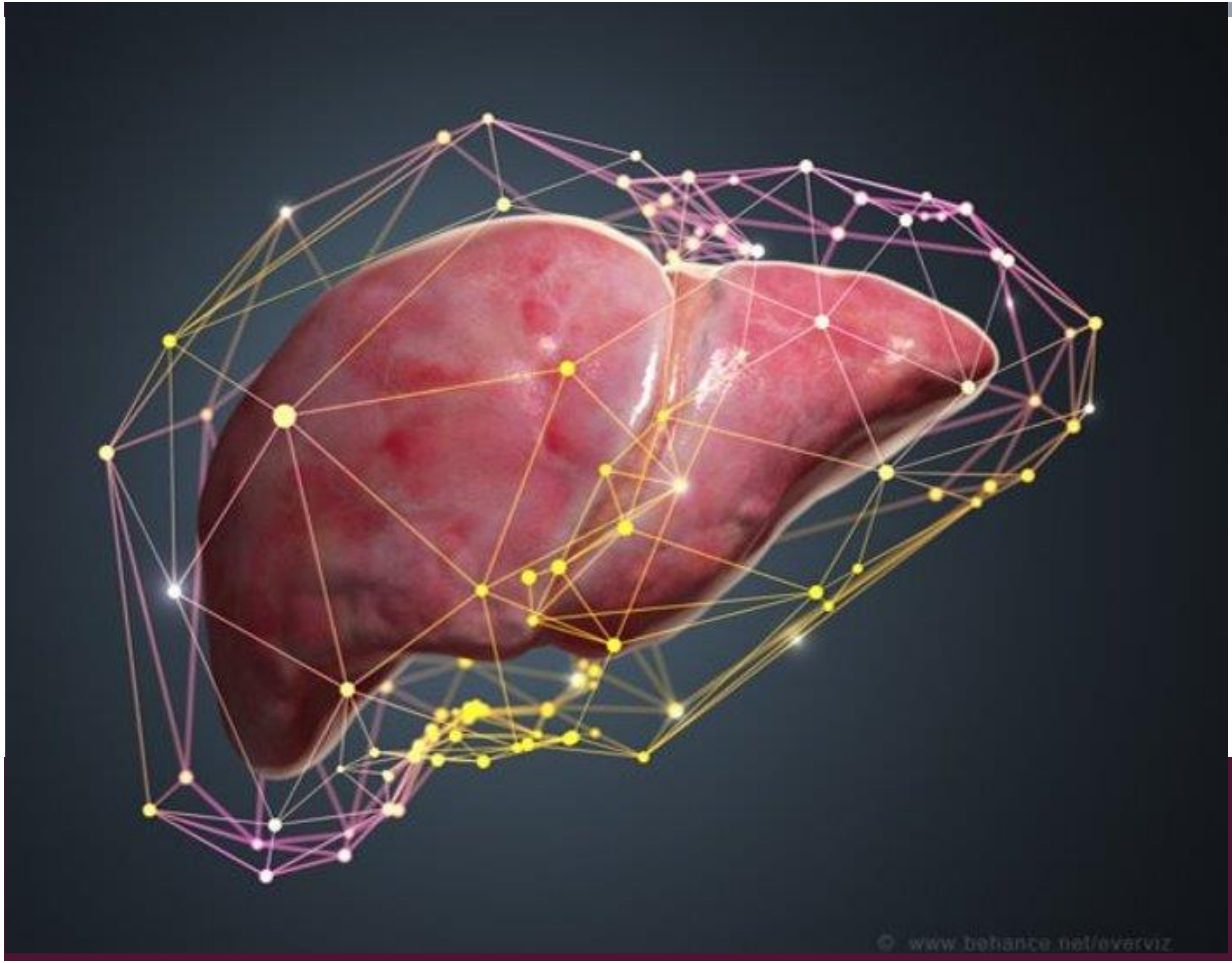
FEASIBILITY OF A MACHINE-LEARNING MODEL TO REDUCE FUTILE PROCUREMENTS IN DONATIONS AFTER CIRCULATORY DEATH IN LIVER TRANSPLANTATION

Results:

- ❑ LightGBM had AUC of 0.833, 0.801 and 0.805 at 30, 45 and 60 mins post-extubation, outperformed DCD-N score and the Colorado Calculator
- ❑ LightGBM had lower futile procurement rates compared to surgeon predictions(0·195 vs 0·078, respectively) and similar missed opportunity rates (0·155 vs 0·167)



Yanagawa R, Iwadoh K, Nakayama T, et al. The Lancet Digital Health, Nov.2025

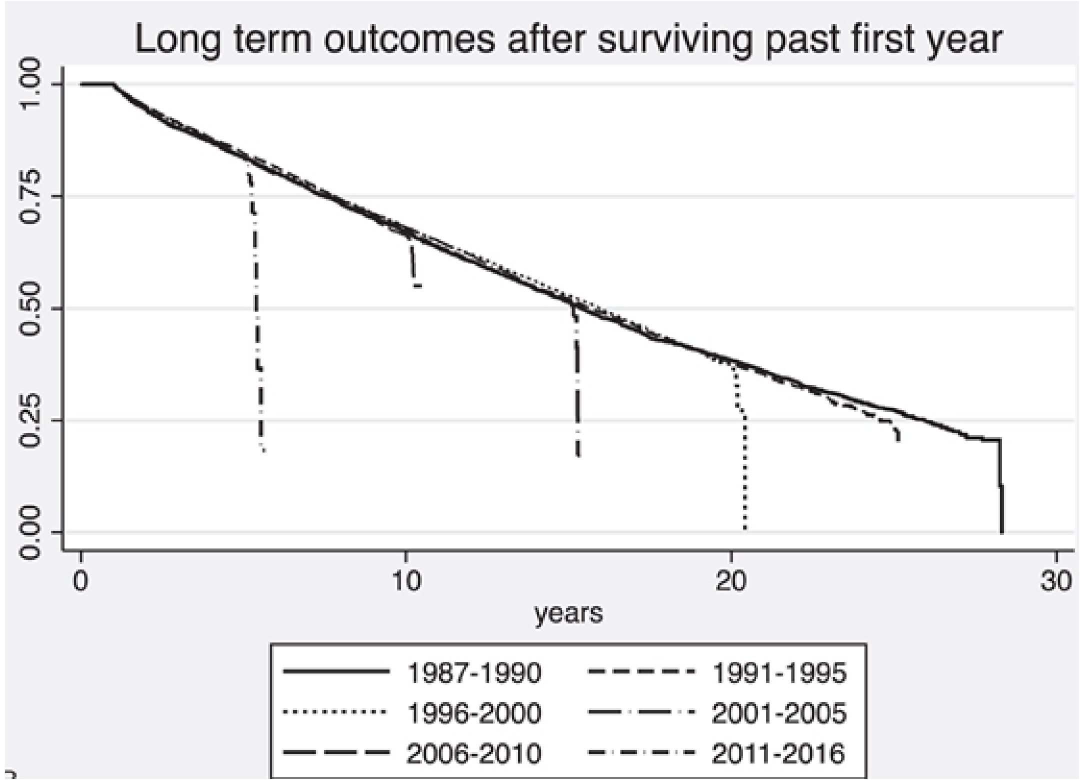


OPTIMIZING POST-TRANSPLANT PATIENT SURVIVAL

POST-TRANSPLANT

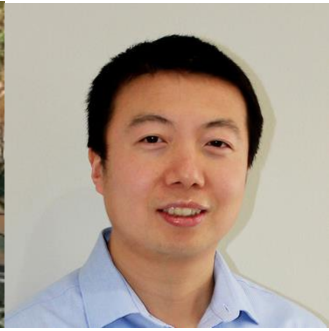
- Median survival of LT recipients 20 years less than population average life expectancy
- Compromised by heightened incidence of cancer, cardiometabolic disease, and infection
- Existing comorbidities, lifelong immunosuppression
- Care of LT recipients based on limited evidence

- **How can we change this and turn Transplant into a Cure?**



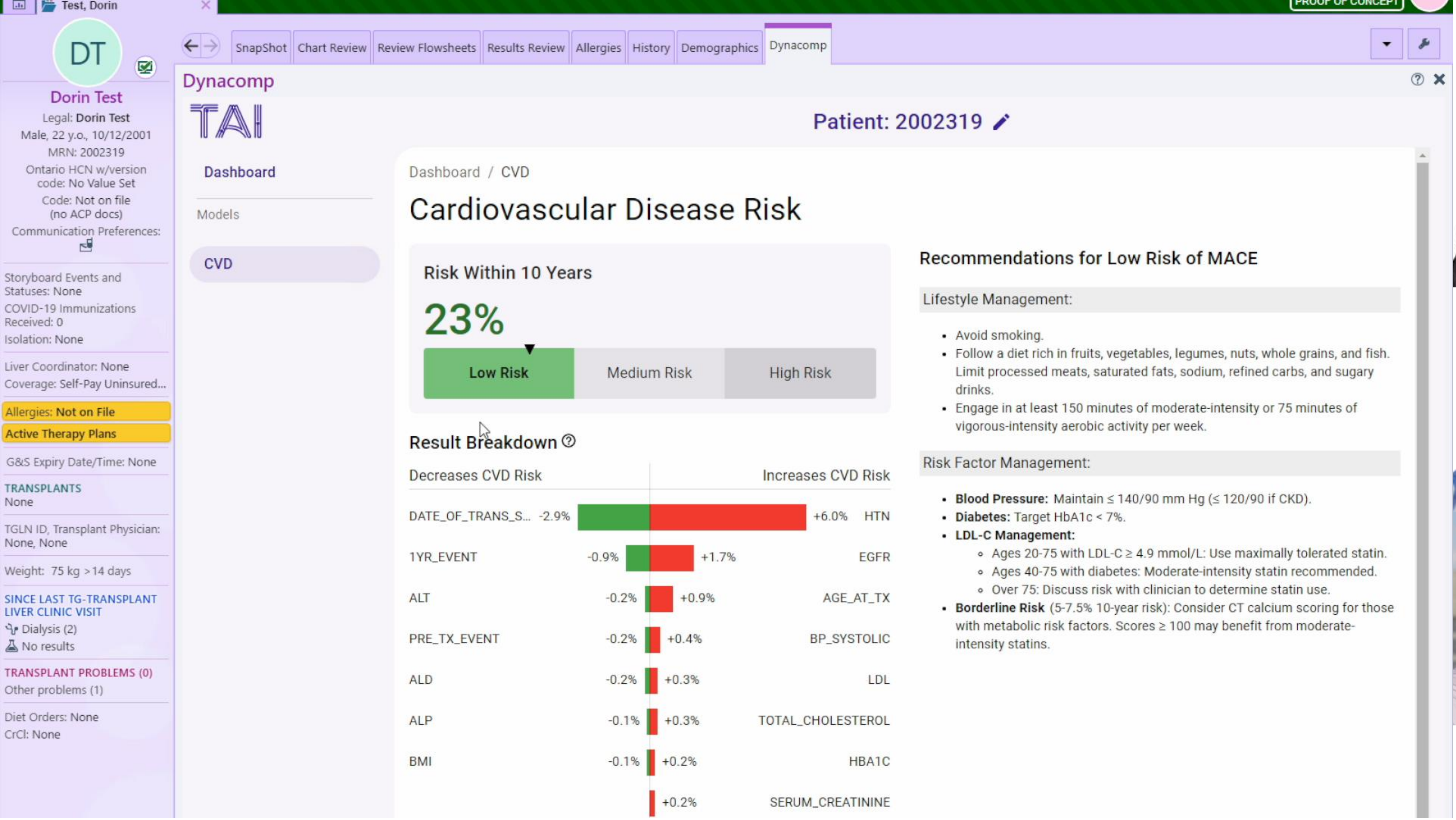
Rana A et al, Annals of Surgery 2019

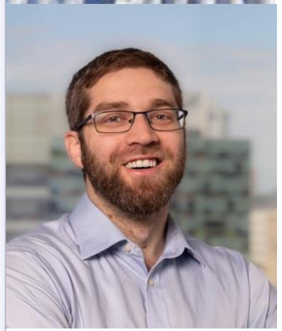
OPTIMIZING LONG-TERM SURVIVAL

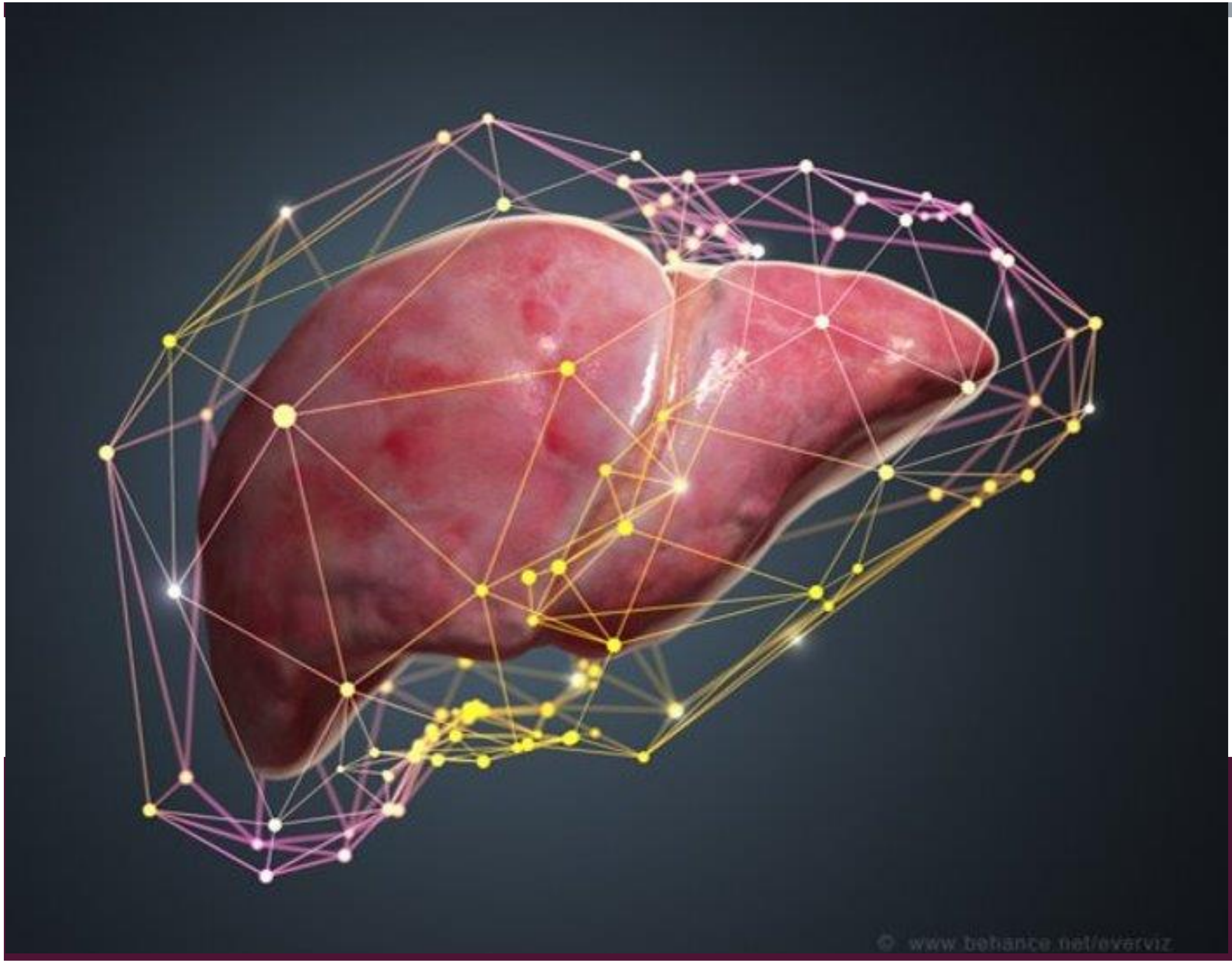


- **Annual** follow-ups, **longitudinal data**
  - Overall Survival
  - Survival compromised by Graft Failure
  - Survival compromised by Infection
  - Survival compromised by Cardiac event
  - Survival compromised by Cancer
- Make a more ‘**personalized**’ risk calculator for managing long term transplant care that can be used at each follow up

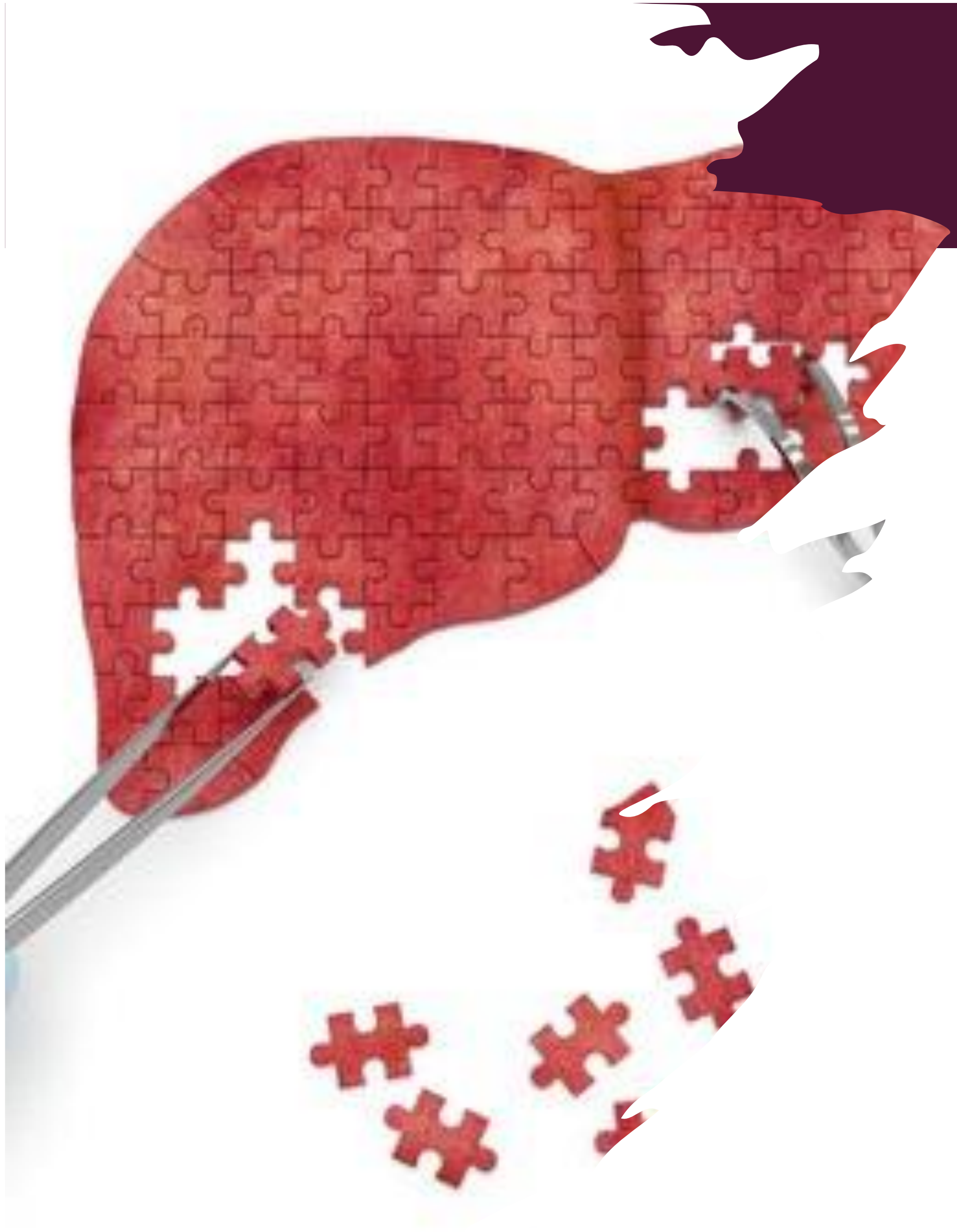
Nitski O, Azhie A, Qazi F, Bhat M*, Wang B*, Lancet Digital Health May 2021


Test, Dorin
PROOF OF CONCEPT
DT
Dorin Test
Legal: Dorin Test
Male, 22 y.o., 10/12/2001
MRN: 2002319
Ontario HCN w/version code: No Value Set
Code: Not on file (no ACP docs)
Communication Preferences:
Storyboard Events and Statuses: None
COVID-19 Immunizations Received: 0
Isolation: None
Liver Coordinator: None
Coverage: Self-Pay Uninsured...
Allergies: Not on File
Active Therapy Plans
G&S Expiry Date/Time: None
TRANSPLANTS
None
TGLN ID, Transplant Physician: None, None
Weight: 75 kg >14 days
SINCE LAST TG-TRANSPLANT LIVER CLINIC VISIT
Dialysis (2)
No results
TRANSPLANT PROBLEMS (0)
Other problems (1)
Diet Orders: None
CrCl: None
SnapShot
Chart Review
Review Flowsheets
Results Review
Allergies
History
Demographics
Dynacomp
Dynacomp
TAI
Patient: 2002319
Dashboard
Models
CVD
Dashboard / CVD
Cardiovascular Disease Risk
Risk Within 10 Years
23%
Low Risk
Medium Risk
High Risk
Result Breakdown
Decreases CVD Risk
Increases CVD Risk
DATE_OF_TRANS_S... -2.9% +6.0% HTN
1YR_EVENT -0.9% +1.7% EGFR
ALT -0.2% +0.9% AGE_AT_TX
PRE_TX_EVENT -0.2% +0.4% BP_SYSTOLIC
ALD -0.2% +0.3% LDL
ALP -0.1% +0.3% TOTAL_CHOLESTEROL
BMI -0.1% +0.2% HBA1C
+0.2% SERUM_CREATININE
Recommendations for Low Risk of MACE
Lifestyle Management:
Avoid smoking.
Follow a diet rich in fruits, vegetables, legumes, nuts, whole grains, and fish. Limit processed meats, saturated fats, sodium, refined carbs, and sugary drinks.
Engage in at least 150 minutes of moderate-intensity or 75 minutes of vigorous-intensity aerobic activity per week.
Risk Factor Management:
Blood Pressure: Maintain ≤ 140/90 mm Hg (≤ 120/90 if CKD).
Diabetes: Target HbA1c < 7%.
LDL-C Management:
Ages 20-75 with LDL-C ≥ 4.9 mmol/L: Use maximally tolerated statin.
Ages 40-75 with diabetes: Moderate-intensity statin recommended.
Over 75: Discuss risk with clinician to determine statin use.
Borderline Risk (5-7.5% 10-year risk): Consider CT calcium scoring for those with metabolic risk factors. Scores ≥ 100 may benefit from moderate-intensity statins.





OPTIMIZING POST-TRANSPLANT GRAFT SURVIVAL



CAUSE OF GRAFT INJURY: DIAGNOSTIC PUZZLE

- Over 50% of LT will have increased Liver enzymes
- Physician has a hunch based on patient profile and test patterns
- <u>Example:</u>
- Young LT recipient
- Elevated LEs (100-200 range)
- ALT>AST and elevated ALP while tapering IS
- Most likely to be Acute rejection

GraftIQ: Multi-class Diagnosis for Graft Pathology

- **Multi-Class ANN** trained on **Longitudinal Clinical, Laboratory features** over week preceding biopsy
- Over 7,000 liver graft biopsies performed at UHN 1987-2024

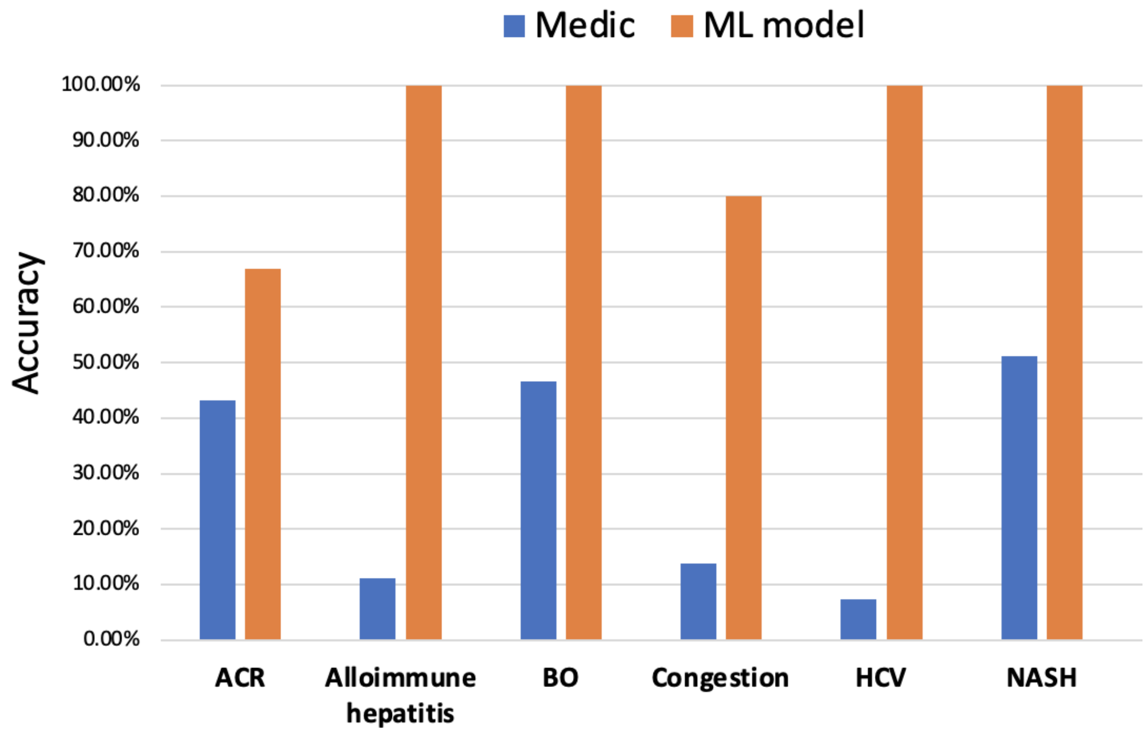
Category	Sensitivity	Specificity	PPV	NPV	AUC
Acute Cellular Rejection	0.8139	0.7366	0.6449	0.8707	0.7752
Alloimmune hepatitis	0.8923	0.9325	0.8021	0.9598	0.9123
Biliary complications	0.9024	0.9112	0.8263	0.9153	0.9021
Vascular Congestion	0.8823	0.9254	0.8155	0.9221	0.9054
Recurrent HCV	0.8630	0.7427	0.7143	0.8791	0.8028
NASH	0.8766	0.8923	0.8119	0.9234	0.9003
Averaged AUC over all the classes : **0.866**					

Performance of ANN in detecting different graft pathologies as well as the overall AUC

Sharma D, Gotlieb N, Bhat M. Nature Communications May 2025

BRINGING THE HUMAN IN THE LOOP TO TEACH MODEL

- **Bayesian inference principles** applied to fuse prior knowledge provided by clinicians
- Integration through **weighted combination of probabilities** by ML model and clinician



Category	AUC Multiclass NN	AUC integrating clinical insight
ACR	0.774	**0.801**
AIH	0.924	0.937
BO	0.902	0.933
Congestion	0.922	0.942
HCV	0.859	0.866
MASLD	0.929	0.930

Sharma D, Gotlieb N, Bhat M. Nature Communications May 2025

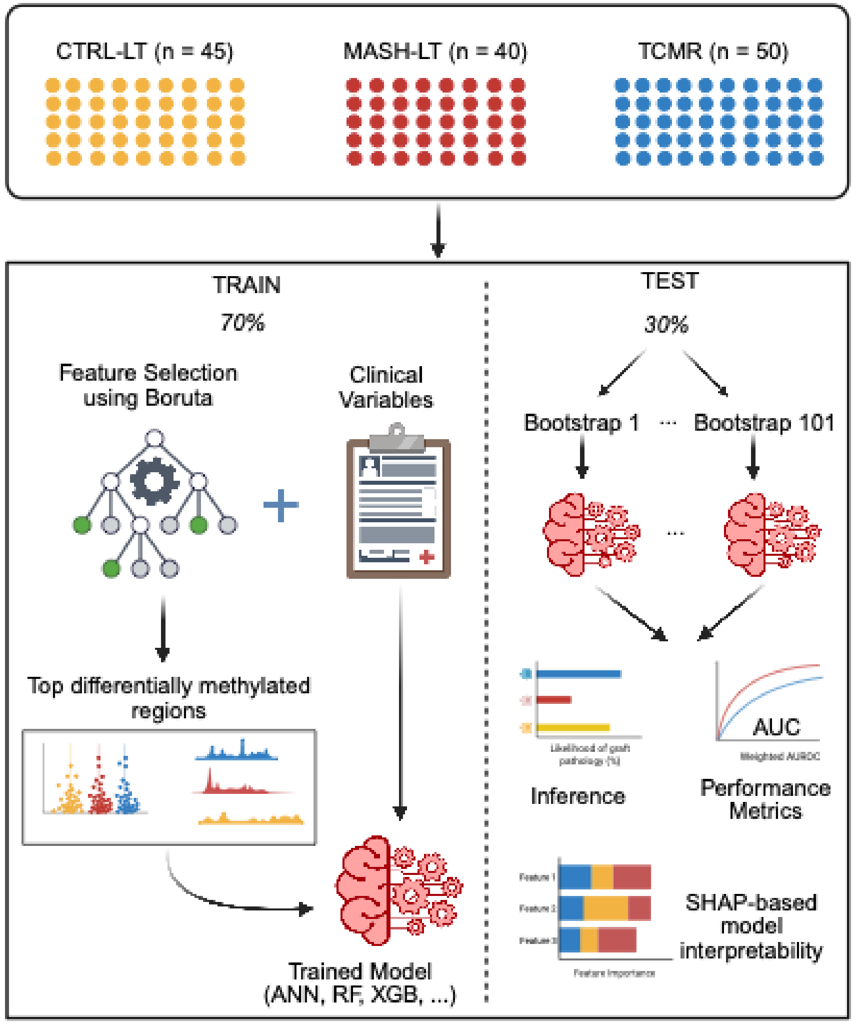
EXTERNAL VALIDATION OF GRAFTIQ

Category	UHN dataset AUC (95% CI)	Mayo dataset AUC (95% CI)	Hannover dataset AUC (95% CI)	NUHS dataset AUC (95% CI)
MASH	0.951[0.939, 0.964]	0.830 (0.785-0.874)	0.722 (0.672, 0.772)	0.737 [0.682, 0.757]
ACR	0.898 [0.886, 0.908]	0.831 (0.805-0.856)	0.754 (0.704, 0.804)	0.698 [0.661, 0.714]
AIH	0.962 [0.949, 0.974]	0.698 (0.578-0.800)	0.708 (0.658, 0.758)	-
BO	0.945 [0.933, 0.956]	0.734 (0.673-0.790)	0.711 (0.647, 0.768)	0.701 [0.665, 0.747]
Congestion	0.923 [0.909, 0.935]	0.716 (0.637-0.787)	-	-
HCV	0.922 [0.909, 0.933]	0.889 (0.867-0.909)	0.768 (0.748, 0.786)	-

Evaluation of the Proposed GraftIQ Model Using Mean AUC and 95% Confidence Intervals on External Validation Datasets. Evaluation results of the proposed GraftIQ model using the mean Area Under the Curve (AUC) metric along with 95% Confidence Intervals (CI) for each diagnostic category, based on four independent external validation datasets.

To further establish the model's robustness, we performed an additional validation on two other international datasets, **Hannover dataset** (n=224) which includes biopsies as follows BO (60.3%), HCV (7.1%), MASH (11.2%), AIH (5.8%), and ACR (15.6%) and NUHS, Singapore dataset with BO (9.6%), MASH (14.4%) and ACR (75.9%). The results with AUCs~0.7 from these datasets as shown in Table 5 columns 4 and 5 in Table 5, confirm the model's reliability, reinforcing its applicability across different medical institutions.

Fine-tuning predictions with Omics data



- TCMR was characterized by enhanced proliferative activity, inflammatory responses
- mTOR signaling and metabolic pathways are key pathways activated in MASH-LT

ML algorithm (CleVER-LG) classifies graft pathology with high accuracy

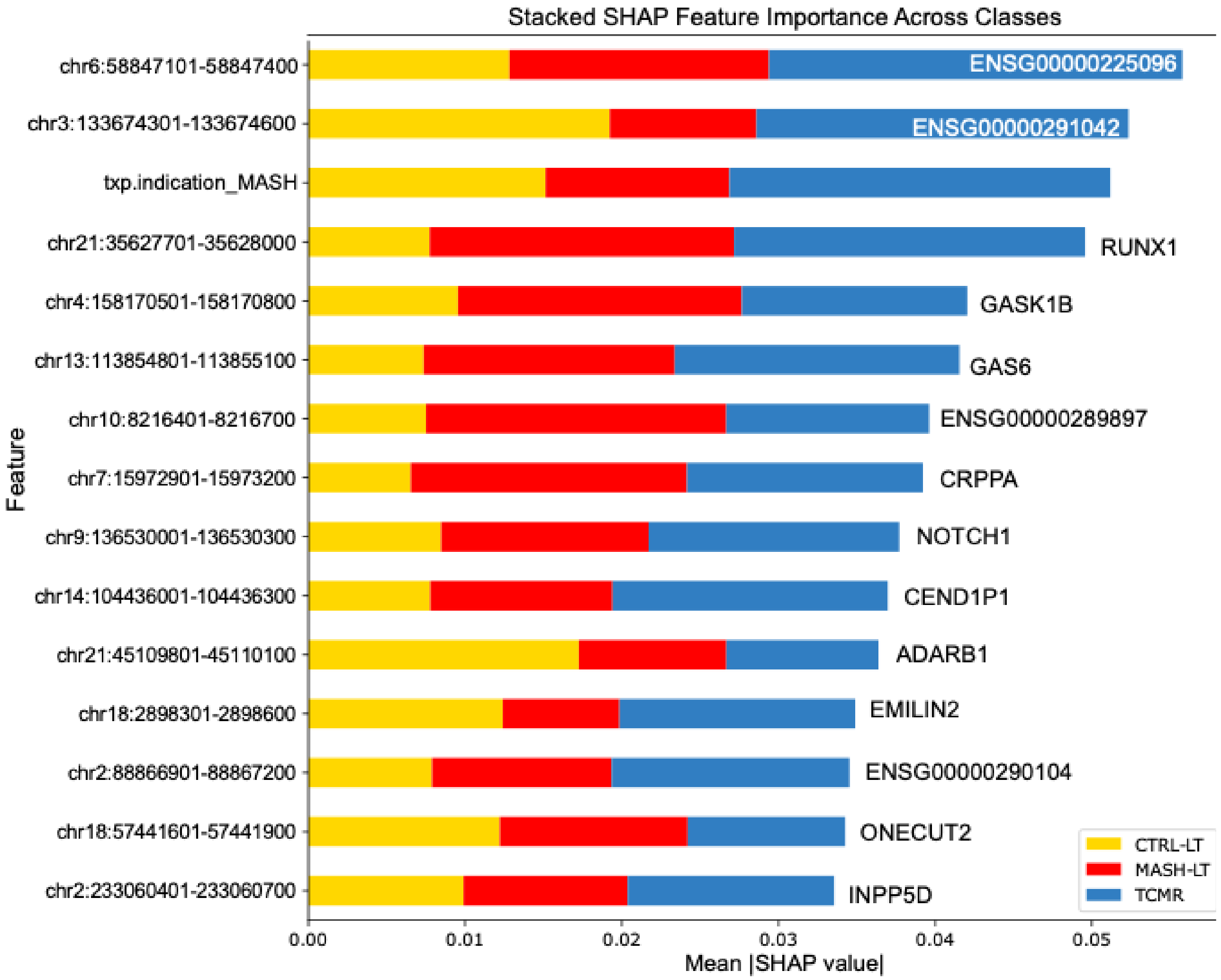
Best Model Performance on Test Set

Artificial Neural Network Model with Dropout Regularization

MODEL TYPE	MODEL NAME	MEAN WEIGHTED AUC (95% CI)	MEDIAN WEIGHTED AUC	MEAN ACCURACY (95% CI)	MEDIAN ACCURACY
COMBINED	ANN	0.911 (0.904 – 0.918)	0.916	0.815 (0.807 – 0.824)	0.820

MEAN PRECISION MACRO (95% CI)	MEDIAN PRECISION MACRO	MEAN RECALL MACRO (95% CI)	MEDIAN RECALL MACRO	MEAN F1 MACRO (95% CI)	MEDIAN F1 MACRO
0.821 (0.809– 0.833)	0.829	0.770 (0.759 – 0.780)	0.769	0.769 (0.757 – 0.781)	0.770

MEAN PPV MACRO (95% CI)	MEDIAN PPV MACRO	MEAN NPV MACRO (95% CI)	MEDIAN NPV MACRO
0.821 (0.809 – 0.833)	0.829	0.920 (0.915 – 0.924)	0.922

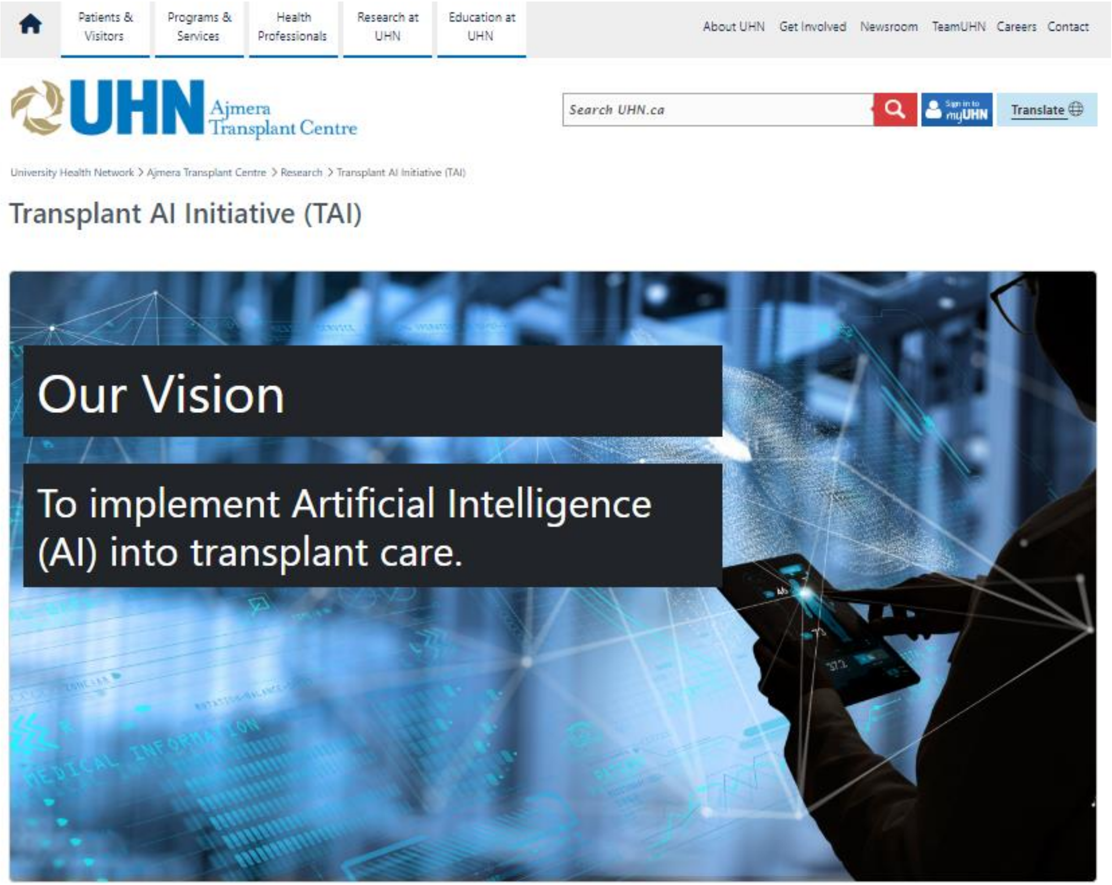


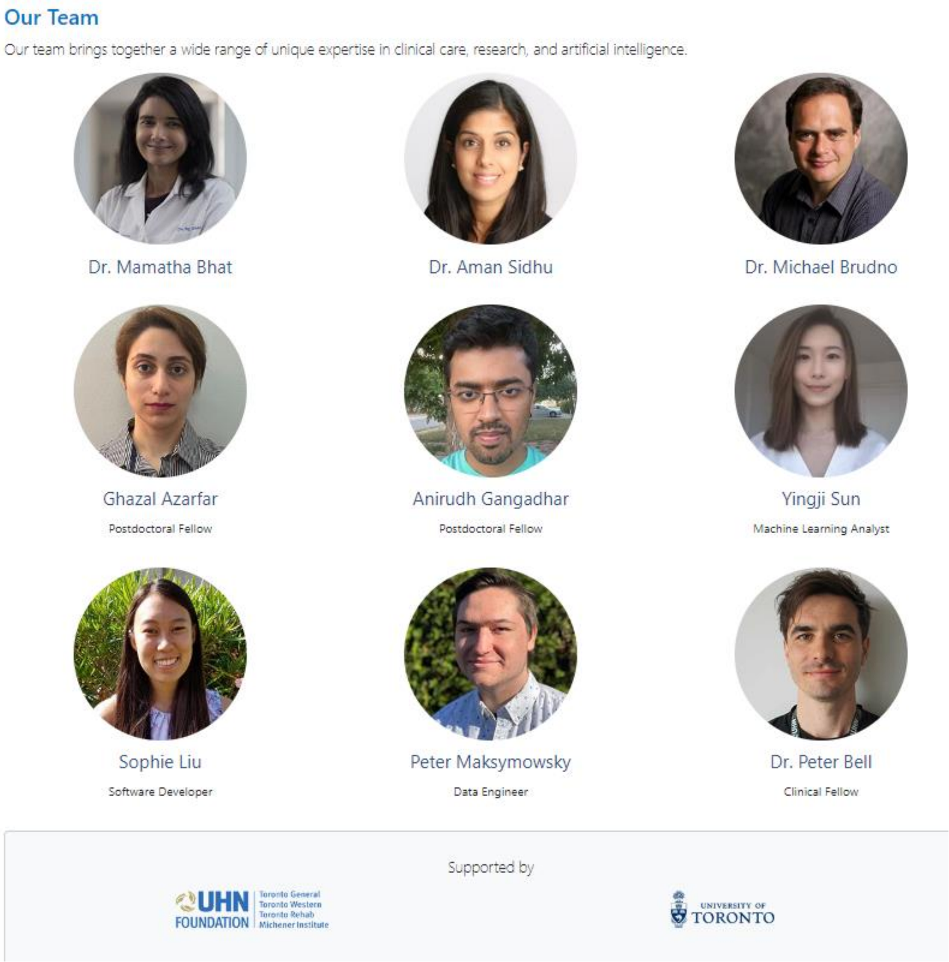
Summary: AI in Transplant

- Complex patient population, with multiple variables impacting outcomes
- Dynamic, complex hidden patterns and relationships
- ML models can be more accurate at **Personalized Predictions**
- **Next Frontier: Multimodal AI, AI Agents, Clinical trials**

Transplant AI Initiative

VISION OF TRANSPLANT AI INITIATIVE (TAI)





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- Layal Jbara (PhD)
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http://bhatlab.ca

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